



Fundamentals of Additive Manufacturing

Groover M.P.: Chapter 29



The student:

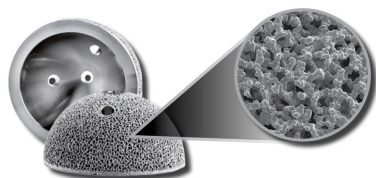
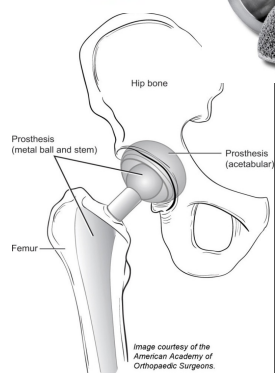
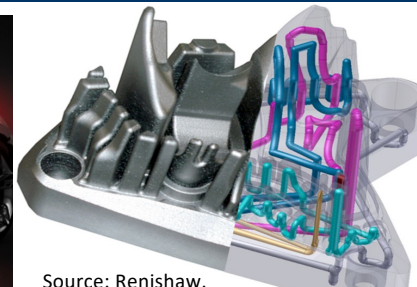
- knows what AM is;
- knows the AM process chain;
- knows the AM process categories.



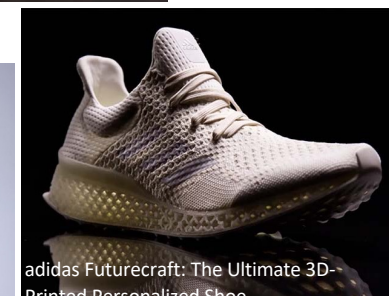
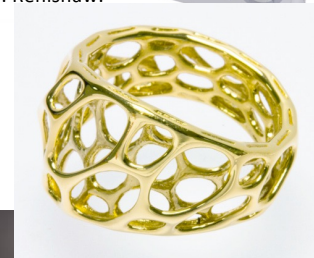
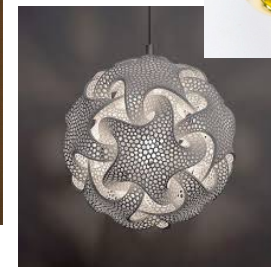


WHAT DO THEY HAVE IN COMMON?

3



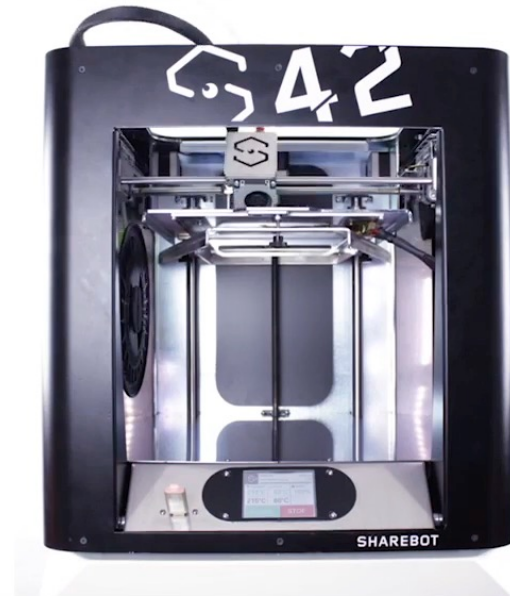
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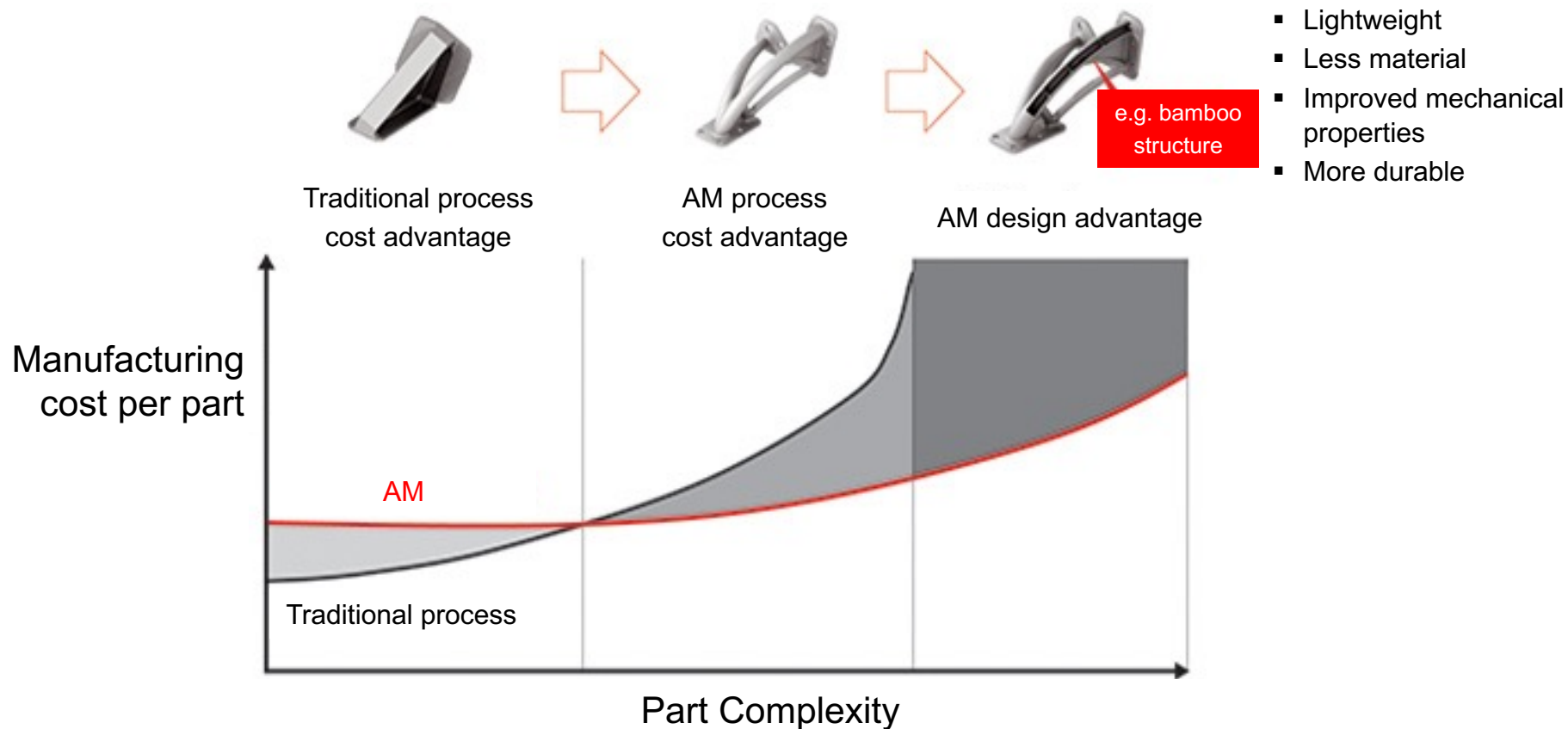
WHAT IS ADDITIVE MANUFACTURING?

Additive manufacturing: “the process of joining materials to make parts from 3D model data, usually layer upon layer, as opposed to subtractive and formative manufacturing methodologies.”





ADDITIVE MANUFACTURING – DESIGN FREEDOM



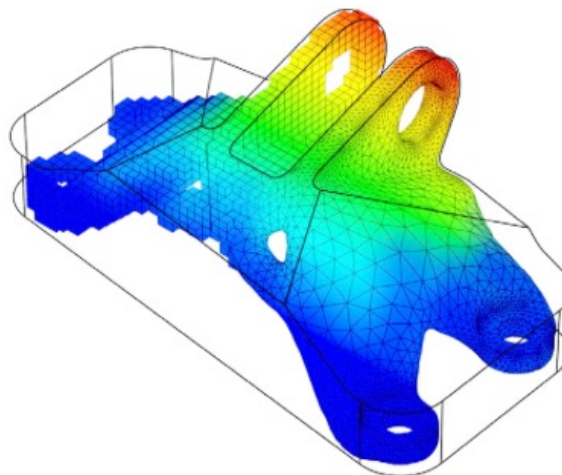


ADDITIVE MANUFACTURING – DESIGN FREEDOM



GE's original design

The original design of the bracket used in the GE Aircraft challenge. The re-designed bracket was required to fit within this volume and meet load requirements while reducing the overall weight of the part.



Cloudmesh optimization

The topology optimized part design generated by Frustum's software.



Final DMP part

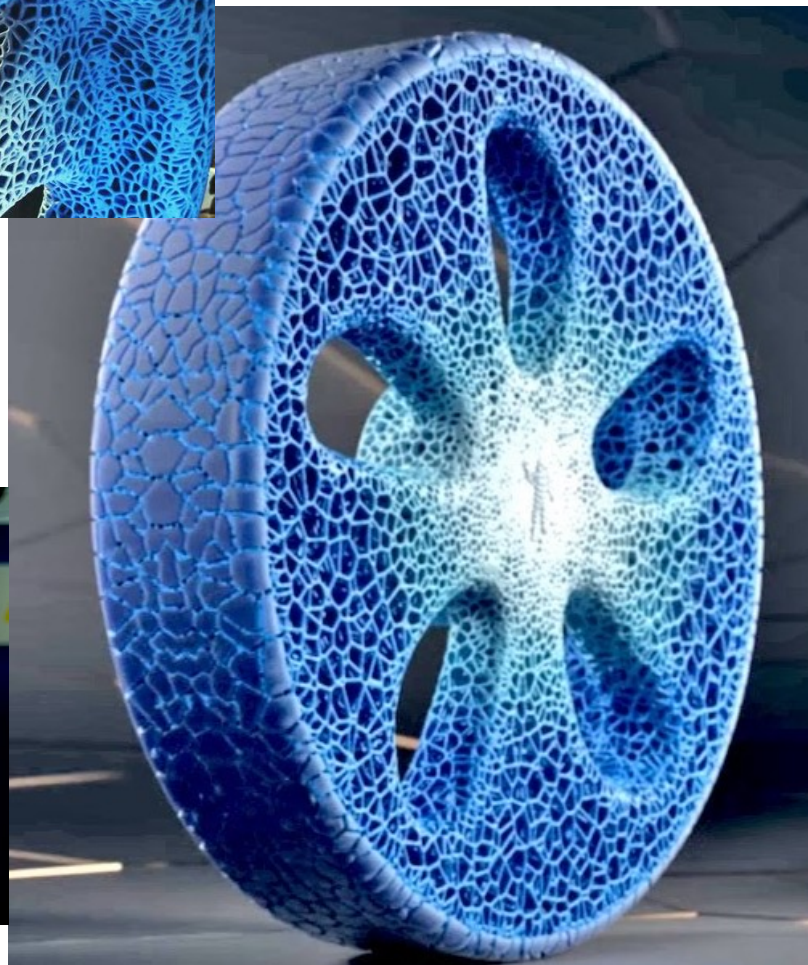
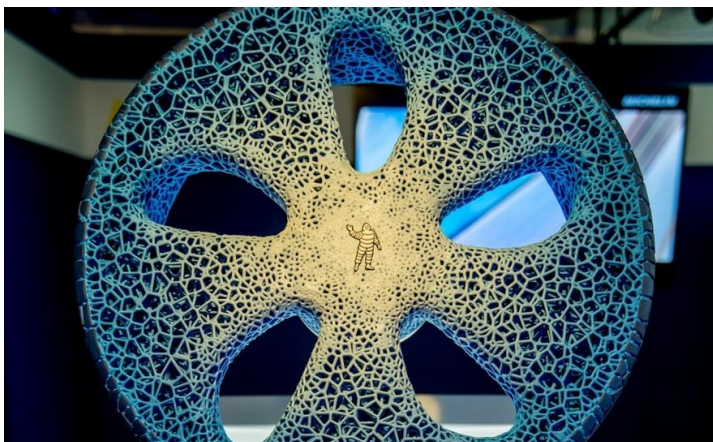
The GE Aircraft engine bracket redesigned by Frustum and manufactured by Quickparts on a 3D Systems ProX DMP 320 printer. The new part passed all the load condition requirements specified by the GE challenge and stayed within the same footprint while reducing weight by a staggering 70 percent.



Michelin has revealed its Vision concept tyre at the 2017 Movin' On conference in Montreal, Canada.

The new visionary Michelin concept tyre is a 3D-printed airless, biodegradable tyre, connected, rechargeable, customizable and organic as well.

(June 2017)

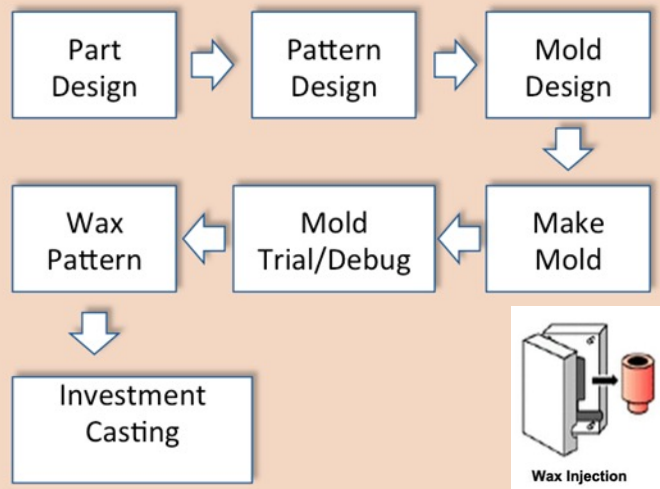




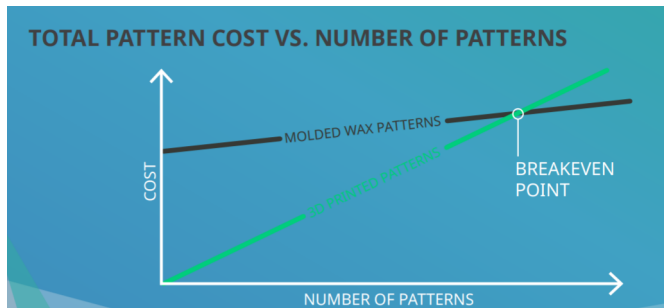
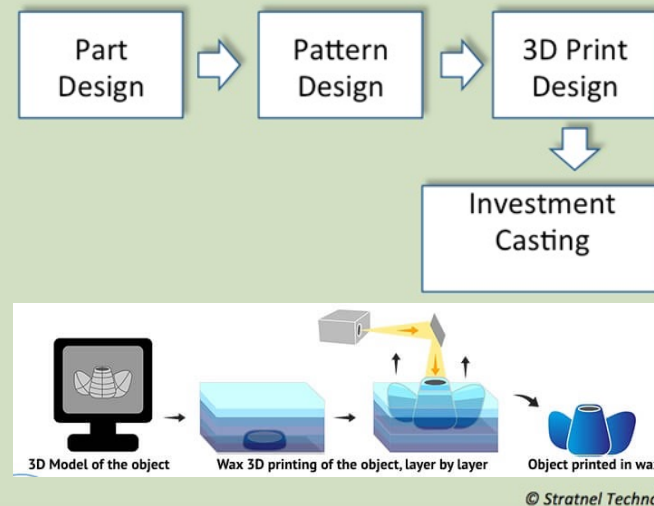
ADDITIVE MANUFACTURING – DESIGN FREEDOM

Rapid Tooling

Investment Casting – Traditional Flow



Investment Casting – 3D Printing Flow



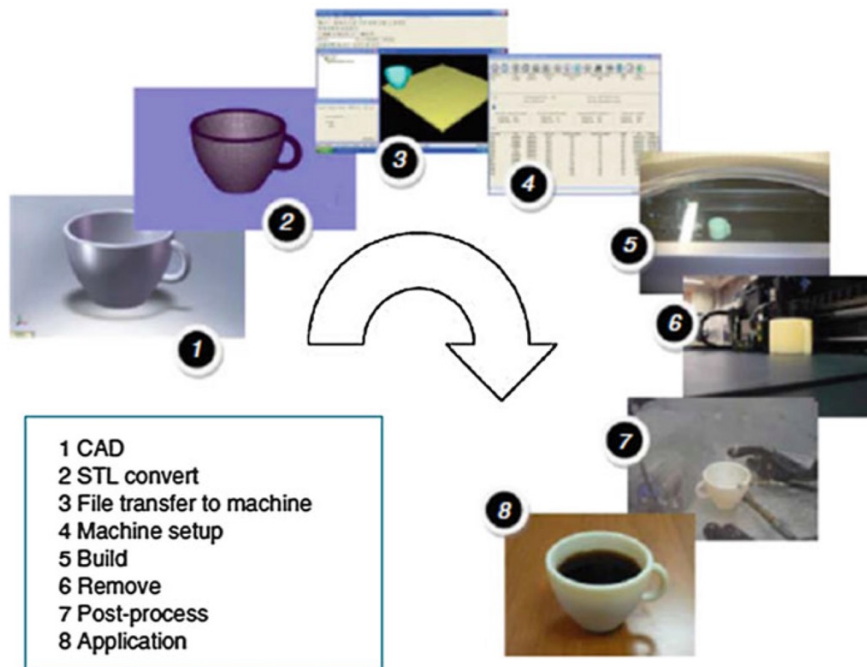
Source: <https://www.3dsystems.com>





AM PROCESS CHAIN

The process chain involved in additive manufacturing technologies is characterised by rapid, flexible and direct fabrication from 3D CAD data. Intermediate stages, such as tool manufacturing, are not necessary.



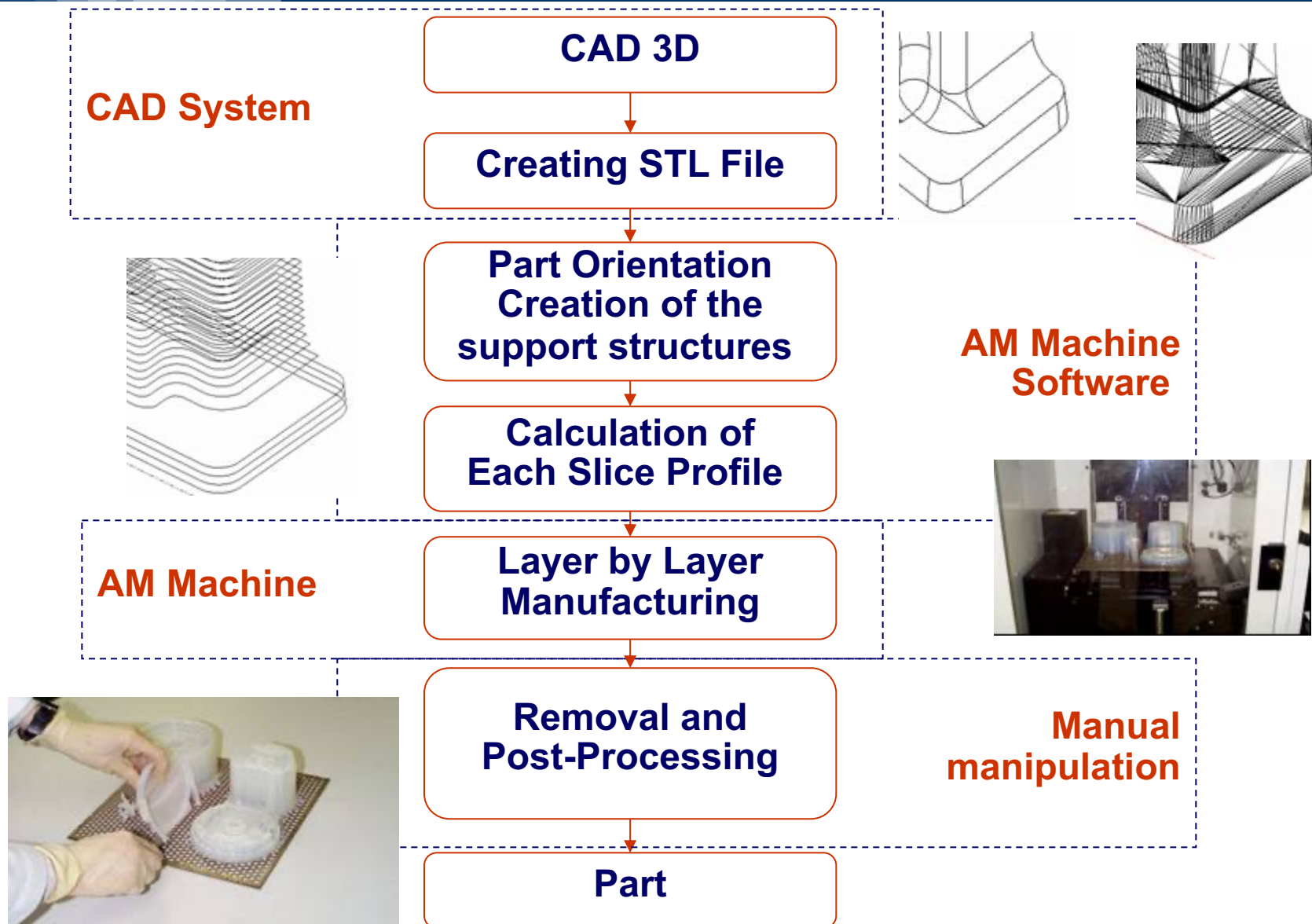
There are basically two different categories:

single-step processes: parts are fabricated in a single operation where the material properties of the intended product are achieved;

multi-step processes: parts are fabricated in two or more operations where the first typically provides the basic geometric shape within the additive manufacturing machine and subsequent post-processing enhances the part's material properties (e.g. using infiltration, sintering, or other thermal or chemical modification).



AM PROCESS CHAIN





According to ISO and ASTM, the AM processes may be grouped in 7 basic categories:

1. Vat photopolymerization
2. Material jetting
3. Binder jetting
4. Powder bed fusion
5. Material extrusion
6. Directed energy deposition
7. Sheet lamination

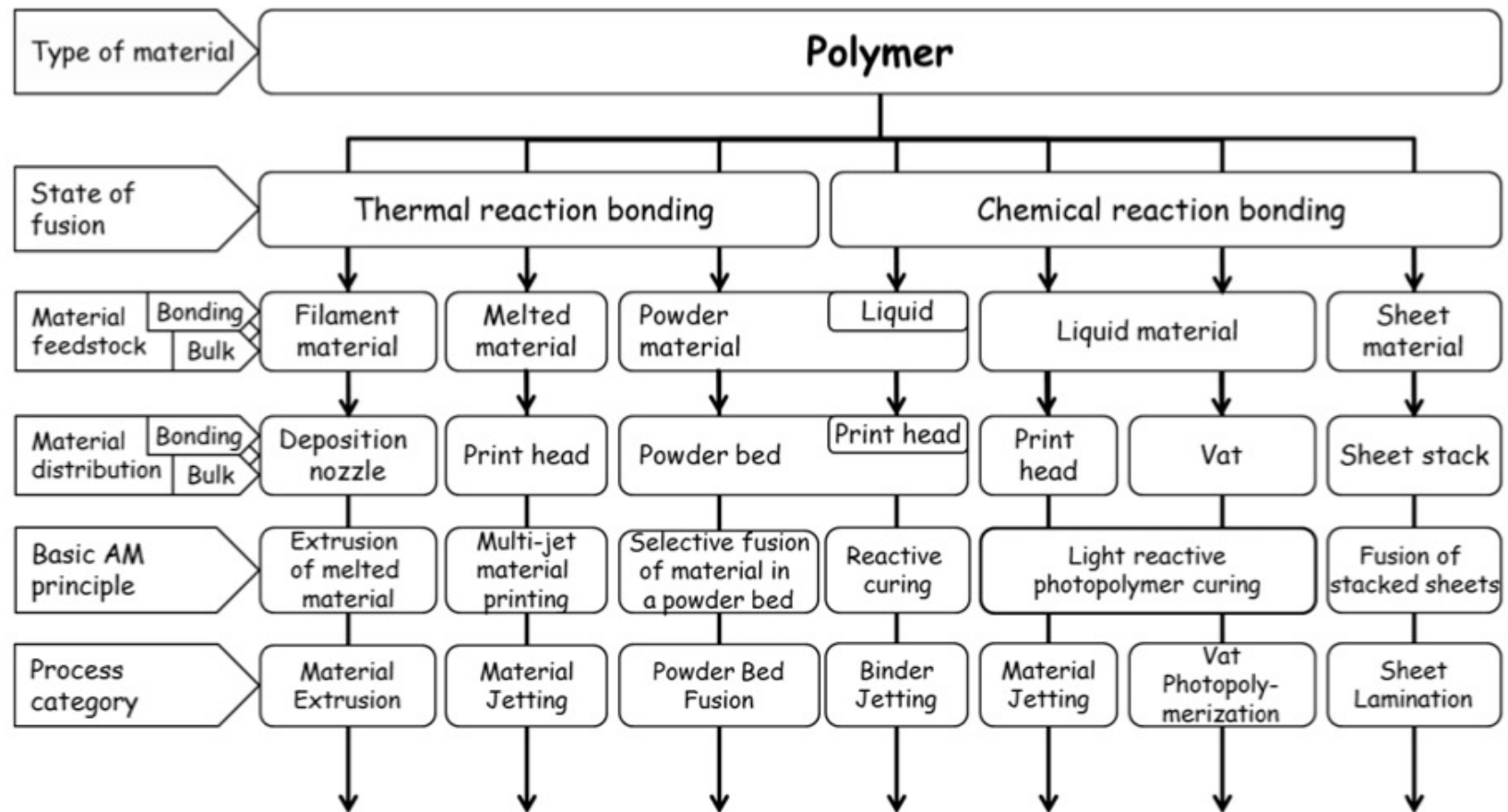


AM PROCESS CATEGORIES AND FEEDSTOCK MATERIALS

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		
Note: Combinations of the above material classes, e.g. a composite, are possible.								



AM PROCESS CATEGORIES



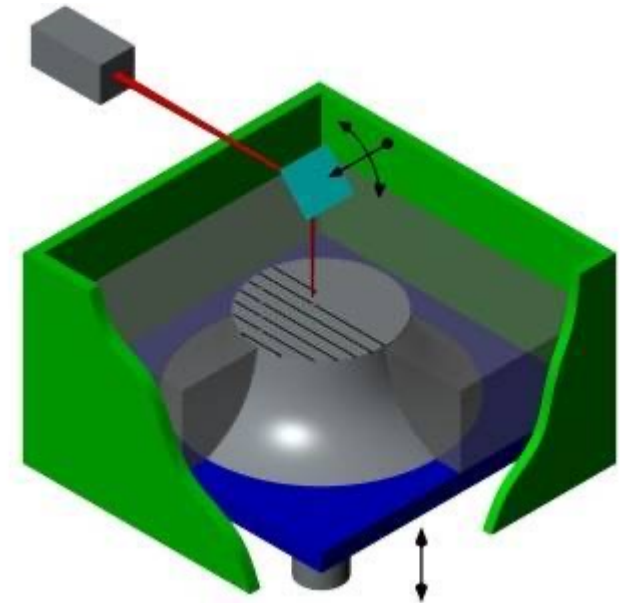


AM PROCESS CATEGORIES

Vat photopolymerization: an additive manufacturing process in which liquid photopolymer in a vat is selectively cured by light-activated polymerization.

Materials	Example materials	Process categories						
		Vat photo-polymeri-zation	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermo-plastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

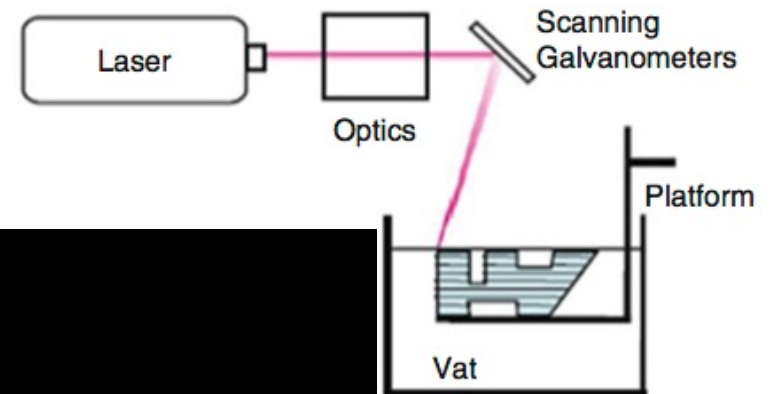
Note: Combinations of the above material classes, e.g. a composite, are possible.



First AM system ever: SLA1 3D System 1987 (Charles (Chuck) Hull)



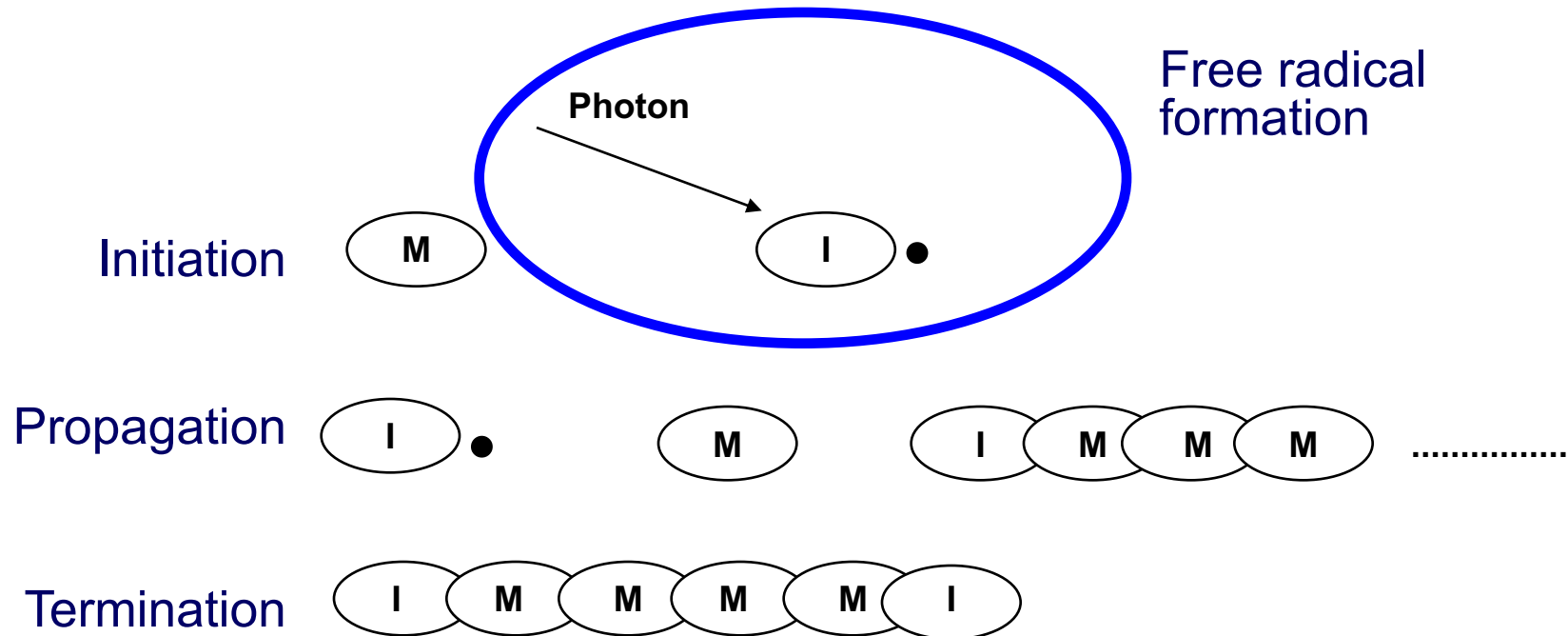
Vector Scan VP Machines: Stereolithography (SL)



<https://youtu.be/NM55ct5Kwil>



Photo-polymerization process



Two main kind:

1. Free-radical photo-polymerization (acrylate)
2. Cationic photo-polymerization (epoxy and vinylether)



Polymerization terminates from one of three causes, recombination, disproportionation, or occlusion.

- **Recombination** occurs when two polymer chains merge by joining two radicals.
- **Disproportionation** involves essentially the cancelation of one radical by another, without joining.
- **Occlusion** occurs when free radicals become “trapped” within a solidified polymer, meaning that reaction sites remain available, but are prevented from reacting with other monomers or polymers by the limited mobility within the polymer network.



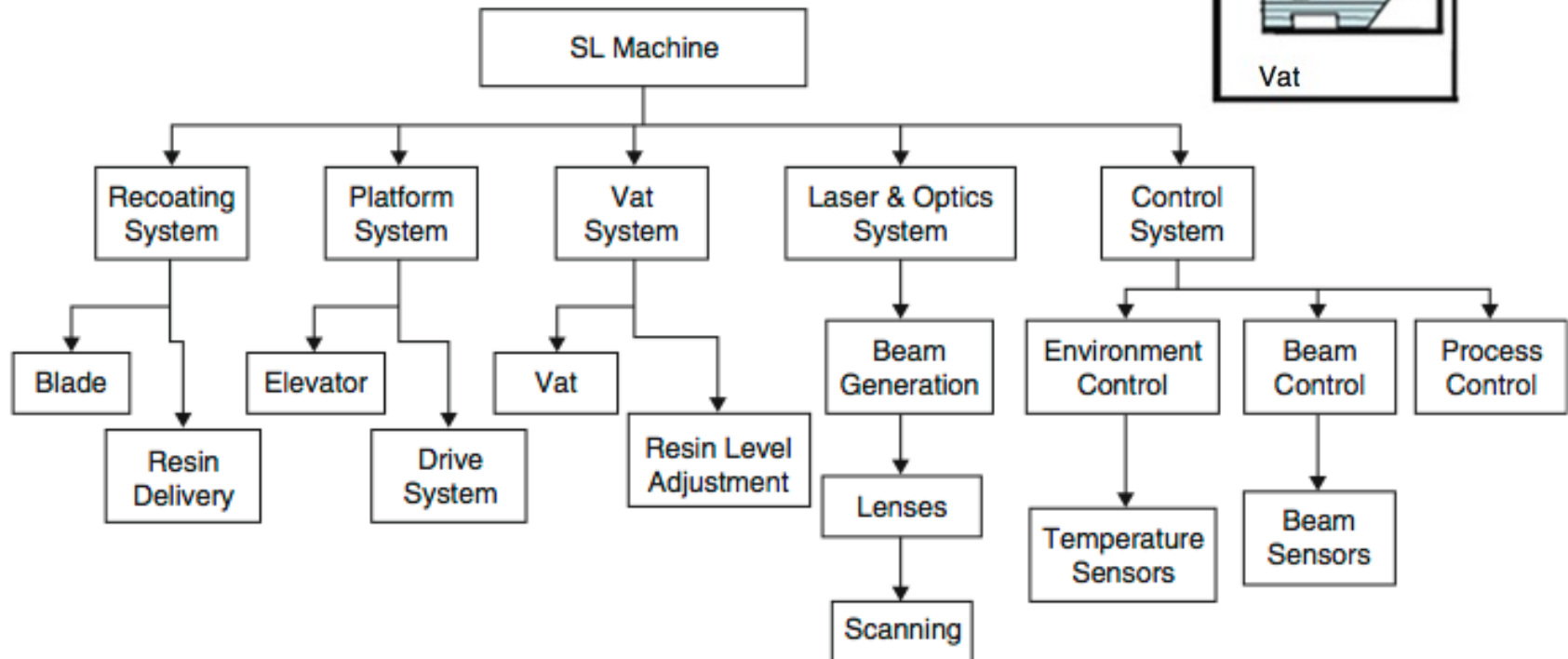
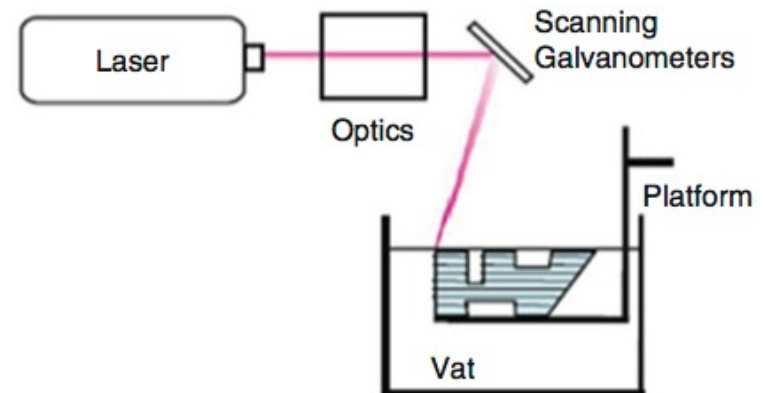
UV curable photopolymers are thermoset polymers:

1. Acrylate-based resins
 - High reactivity (*photospeed*)
 - 5-20% shrinkage → inaccuracy (curl distortion)
 - Susceptible to oxygen inhibition
2. Epoxy resins
 - Low photospeed
 - 1-2% shrinkage
 - Not susceptible to oxygen inhibition
 - Brittleness
3. Commercially available resins: epoxides with some acrylate content. They form a polymer blend rather than a copolymer: monomers undergo independent polymerization.



VAT PHOTOPOLYMERIZATION

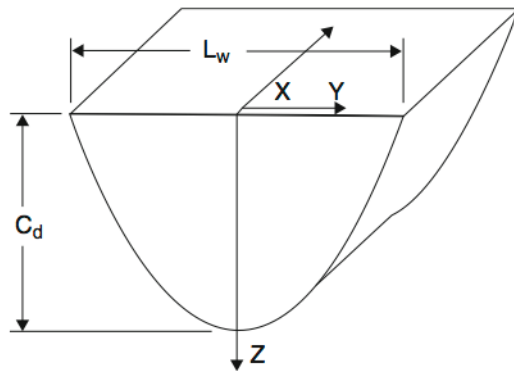
Vector Scan VP Machines: Stereolithography (SL)





VAT PHOTOPOLYMERIZATION

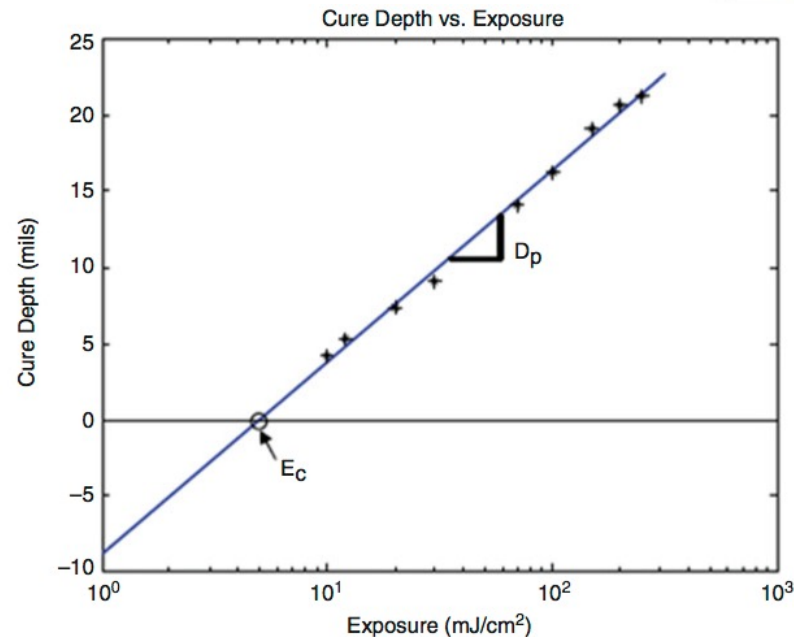
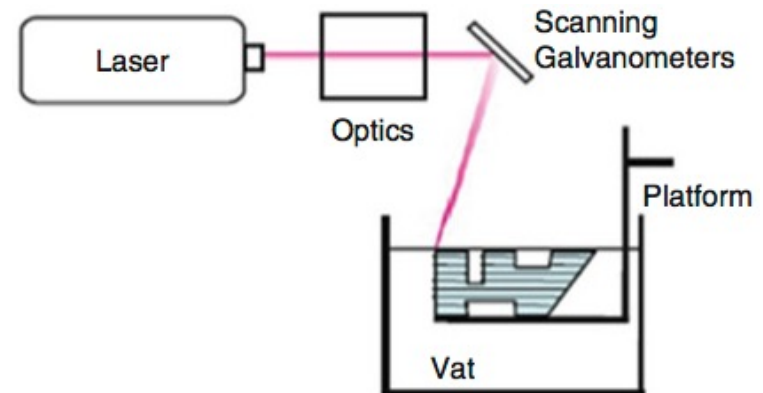
Scan Patterns



Cured line showing parabolic shape

C_d : cure depth,

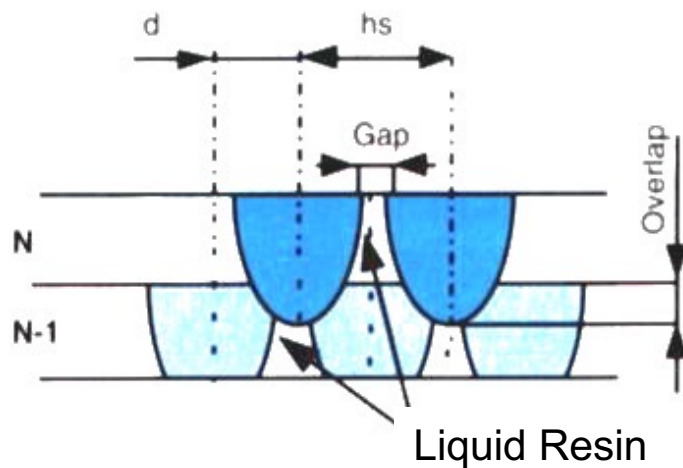
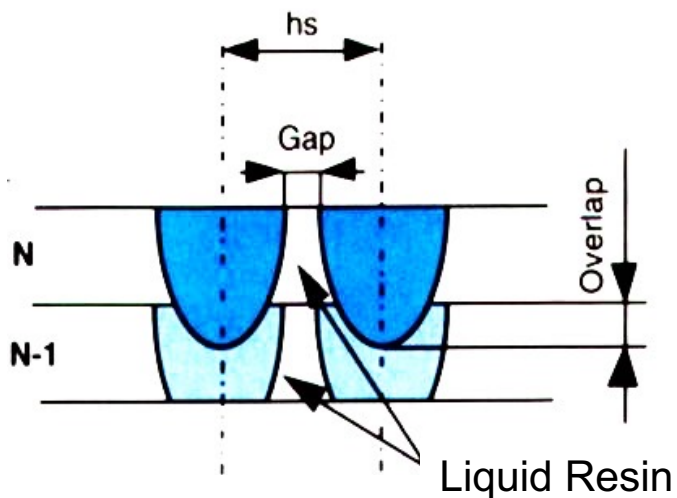
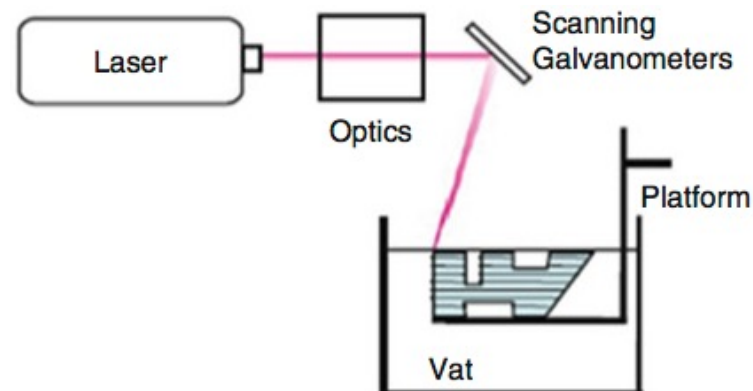
L_w : line width.



Resin “working curve”: cure depth vs. exposure
(1 mil = 0,001 in. = 0,25 mm)

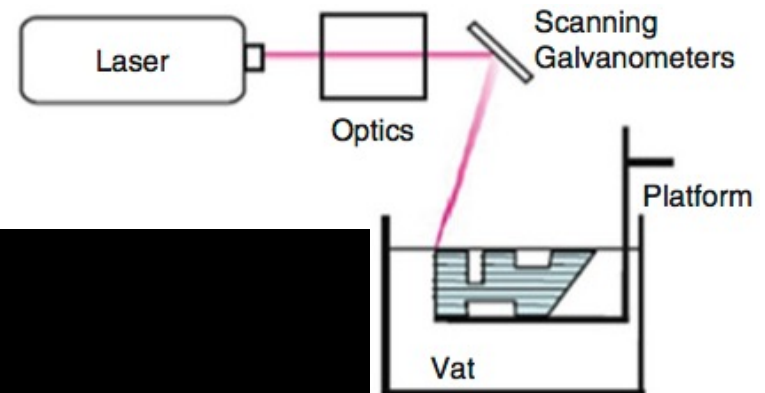


Scan Patterns





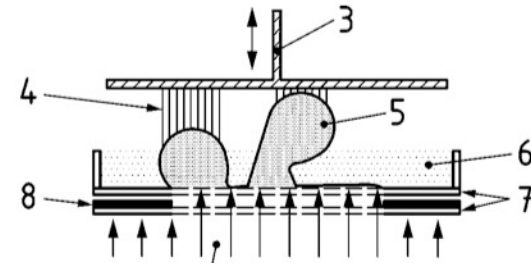
Vector Scan VP Machines: Stereolithography (SL)



<https://youtu.be/V7PeN9F4Dns>



Vector Scan VP Machines: Stereolithography (SL)



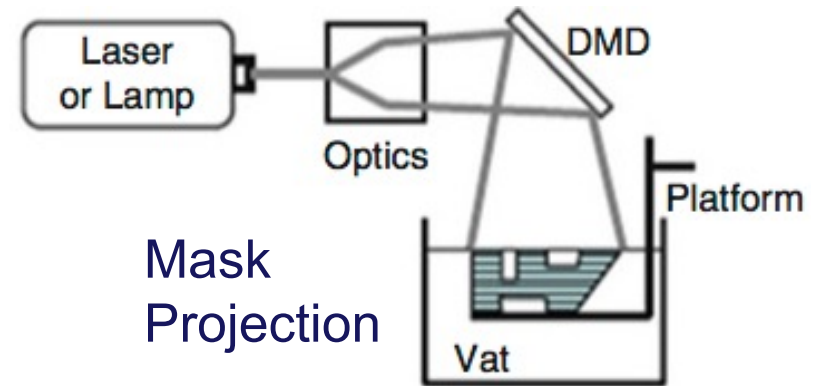
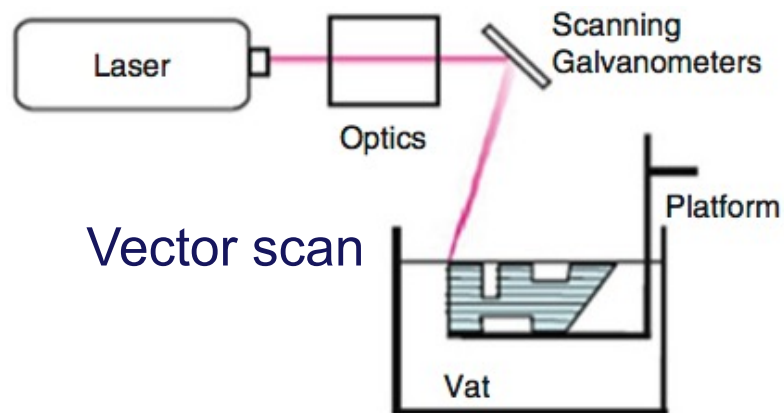
Introduction to Stereolithography

<https://youtu.be/yW4EbCWaJHE?list=RDyW4EbCWaJHE>



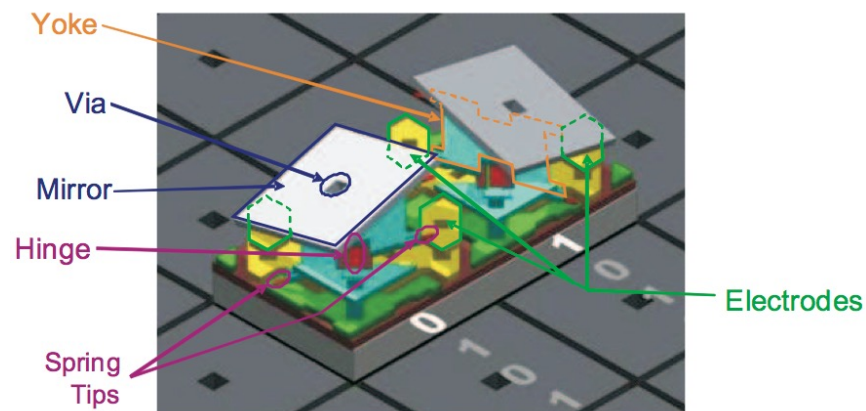
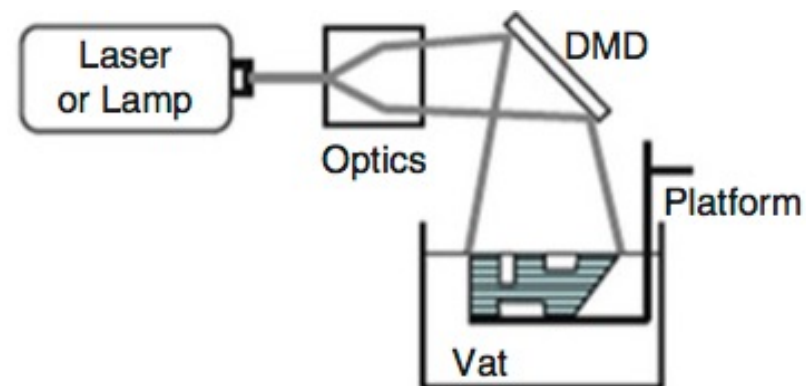
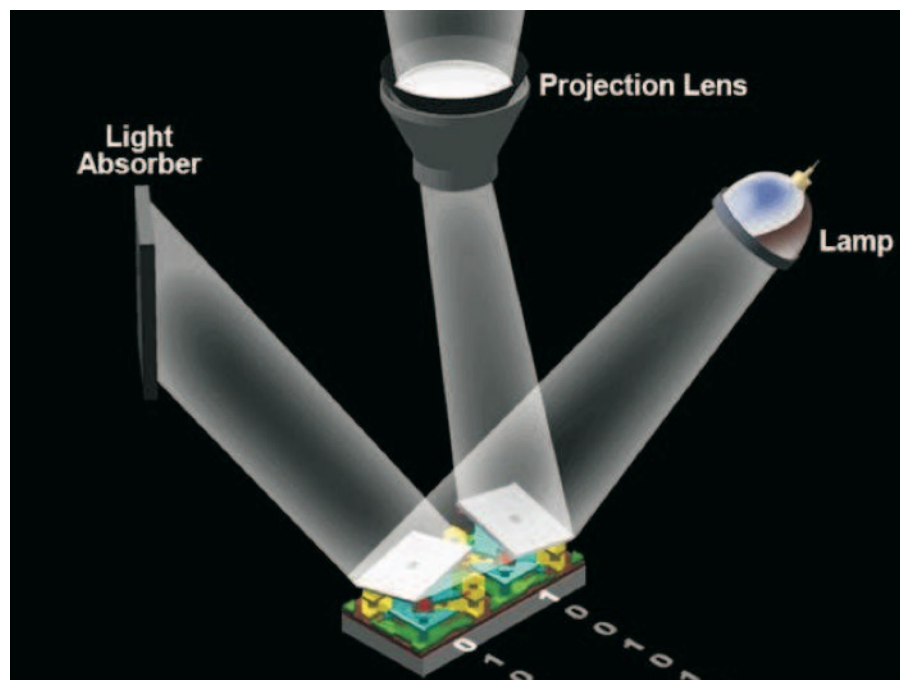
VAT PHOTOPOLYMERIZATION

Schematic diagrams of three approaches to photopolymerization processes.



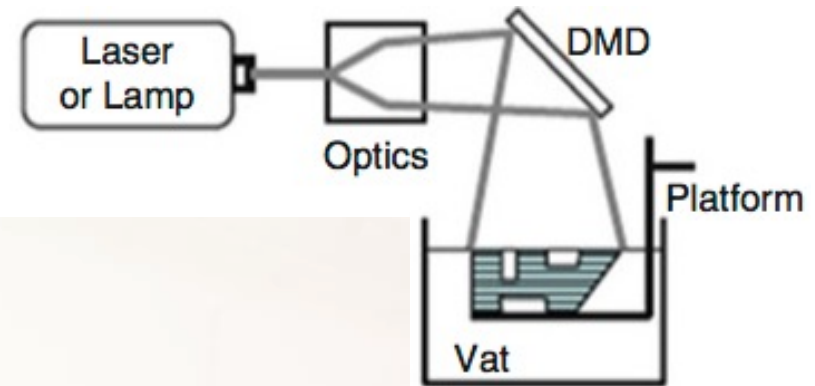


Mask Projection VP Machines: Mask Projection Stereolithography (MPSL)





Mask Projection VP Machines: Mask Projection Stereolithography (MPSL)



<https://youtu.be/l104iZ2w4zg>

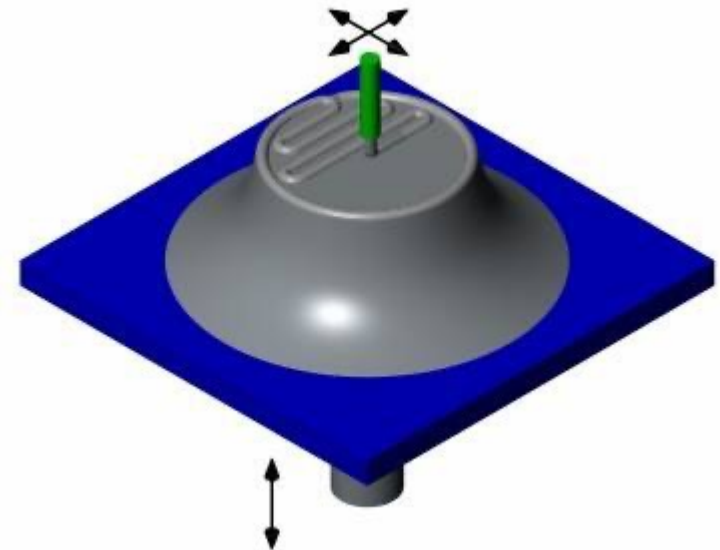


AM PROCESS CATEGORIES

Material extrusion: an additive manufacturing process in which material is selectively dispensed through a nozzle or orifice.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.

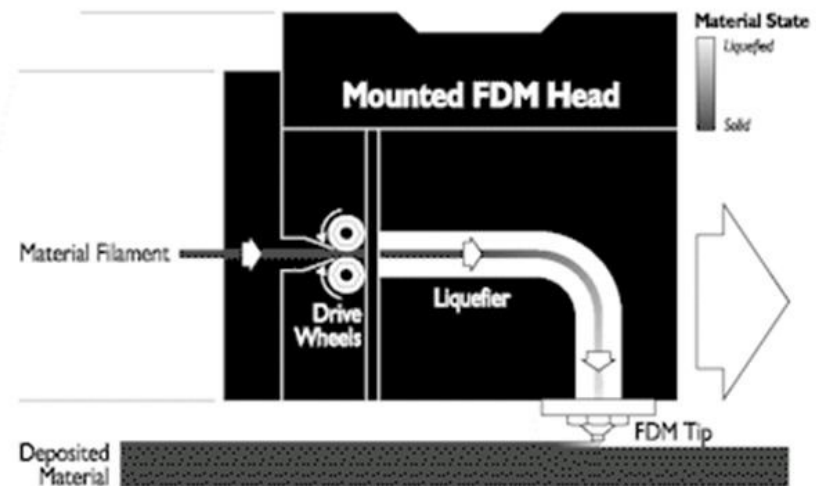
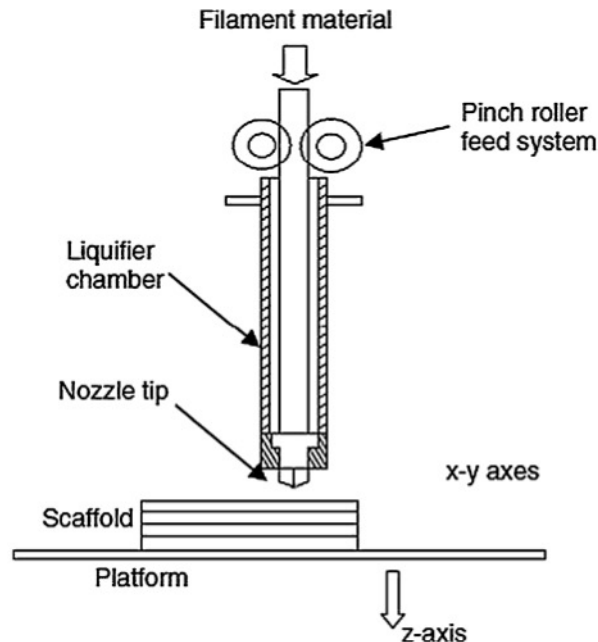
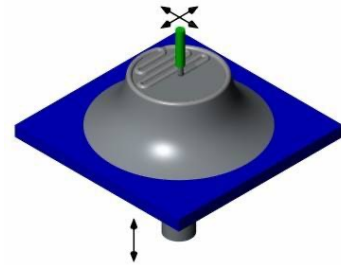




MATERIAL EXTRUSION

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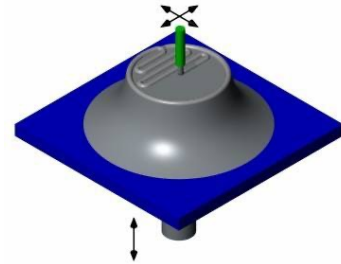
A continuous filament can be pushed into the reservoir chamber, thus providing a mechanism for generating an input pressure for the nozzle.





The machine must be capable of:

- scanning in a horizontal plane (controlling the workhead in x-y plane during each layer);
- starting and stopping the flow of material while scanning;
- moving upwards by a distance equal to one layer in the z-direction (or the part downwards);

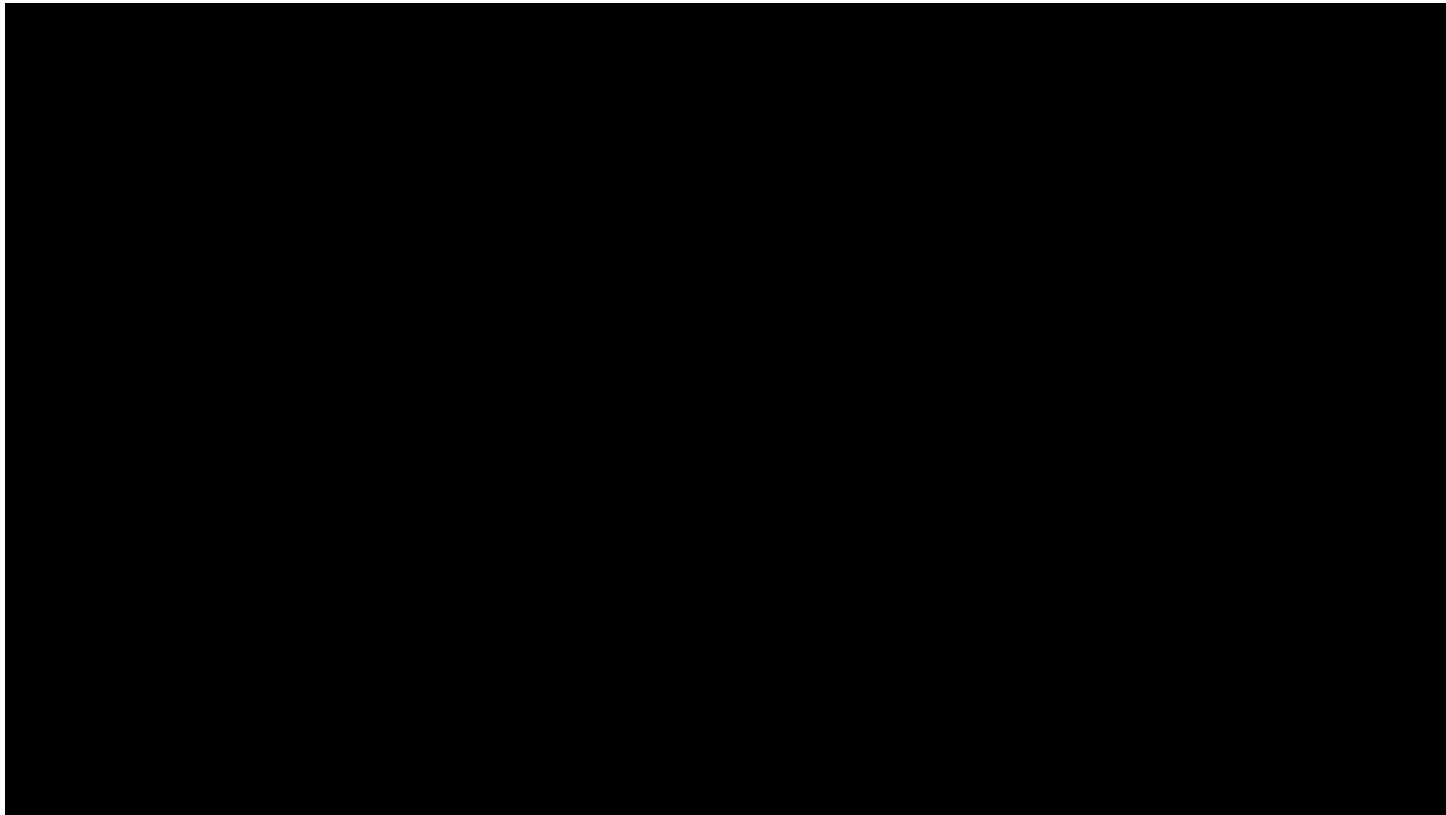


Part is fabricated layer-by-layer from the base up.



MATERIAL EXTRUSION

An additive manufacturing process in which material is selectively dispensed through a nozzle or orifice.



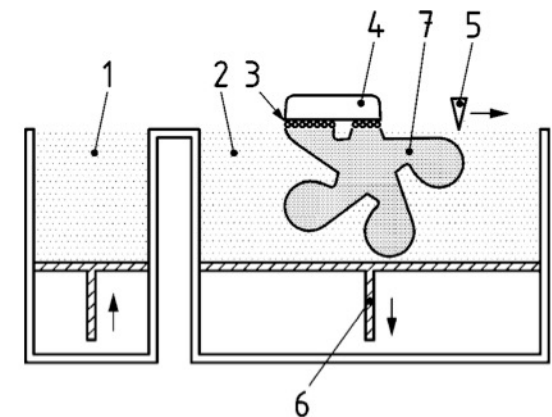
<https://youtu.be/WHO6G67GJbM>



Binder jetting: an additive manufacturing process in which a liquid bonding agent is selectively deposited to join powder materials.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.

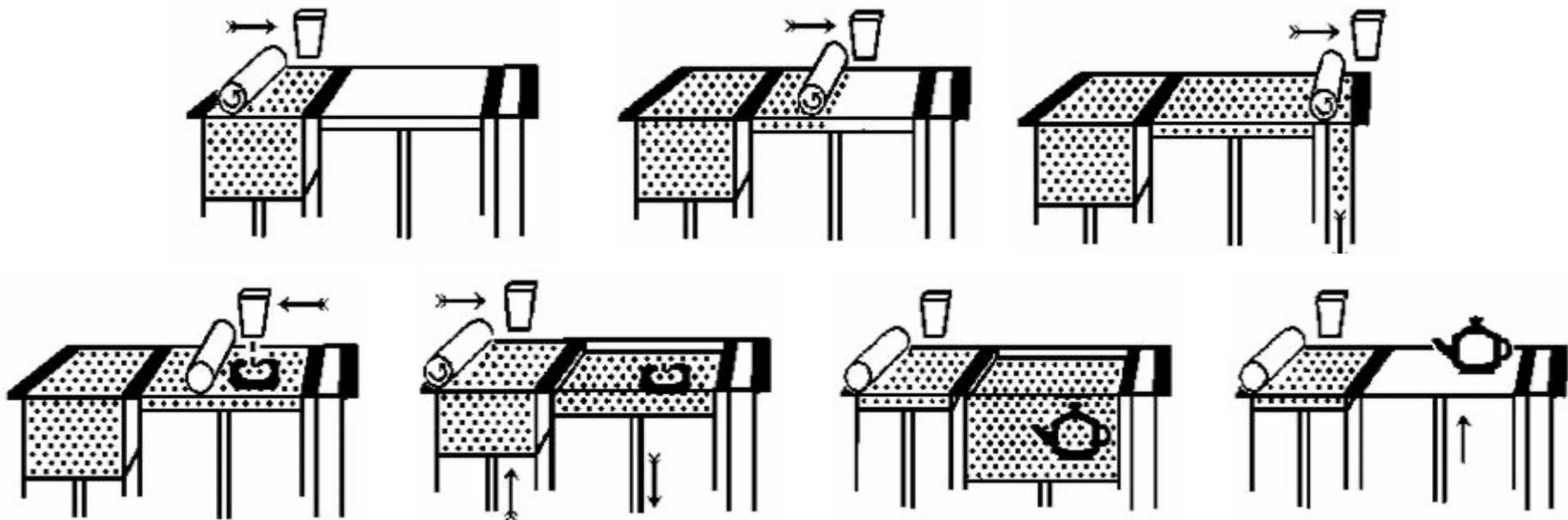


Key

- 1 Powder feeding system
- 2 Powder material distributed in a powder bed
- 3 Liquid bonding agent
- 4 dispensing apparatus including connection to bonding agent feed system
- 5 Powder spreading device
- 6 Build platform and elevator
- 7 Product



Binder jetting methods were developed in the early 1990s, primarily at MIT. They developed what they called the 3D Printing (3DP) process in which a binder is printed onto a powder bed to form part cross sections.





When Z Corporation first started in the mid-1990s, their first material was **starch** based and used a **water-based binder** similar to a standard house-hold glue.

At present, the commercially available powder

- **plaster based** (calcium sulfate hemihydrate) with water-based binder;
- **PMMA** (poly-methyl methacrylate) powder and uses a liquid binder that reacts at room temperature;
- **wax-based binder** for use with PMMA powder to form pattern for investment casting.

With either the starch or polymer powders, parts are typically considered visual prototypes or light-duty functional prototypes.

Infiltrants are used to strengthen the parts after they are removed from the powder bed. In some cases, particularly with elastomeric infiltrants, parts can be used for functional purposes.



<https://youtu.be/L6Rd9dilkrs>

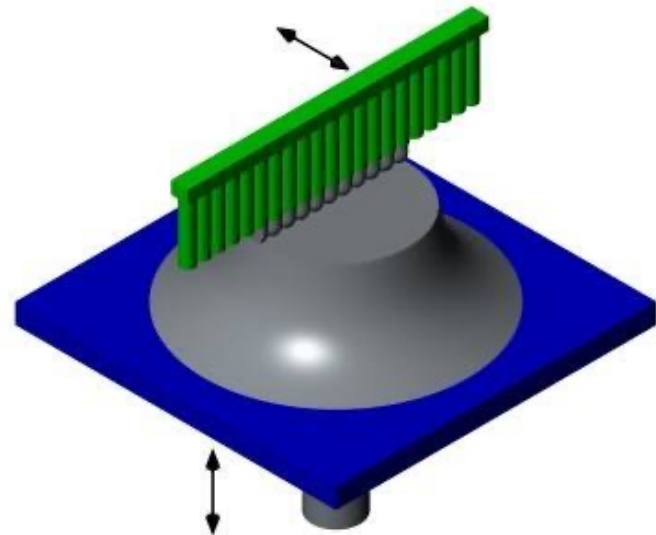


AM PROCESS CATEGORIES

Material jetting: an additive manufacturing process in which droplets of build material are selectively deposited.

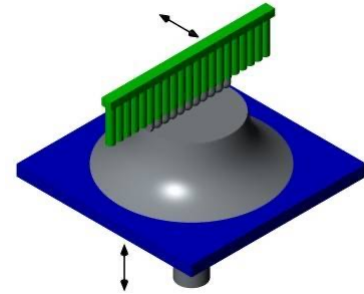
Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



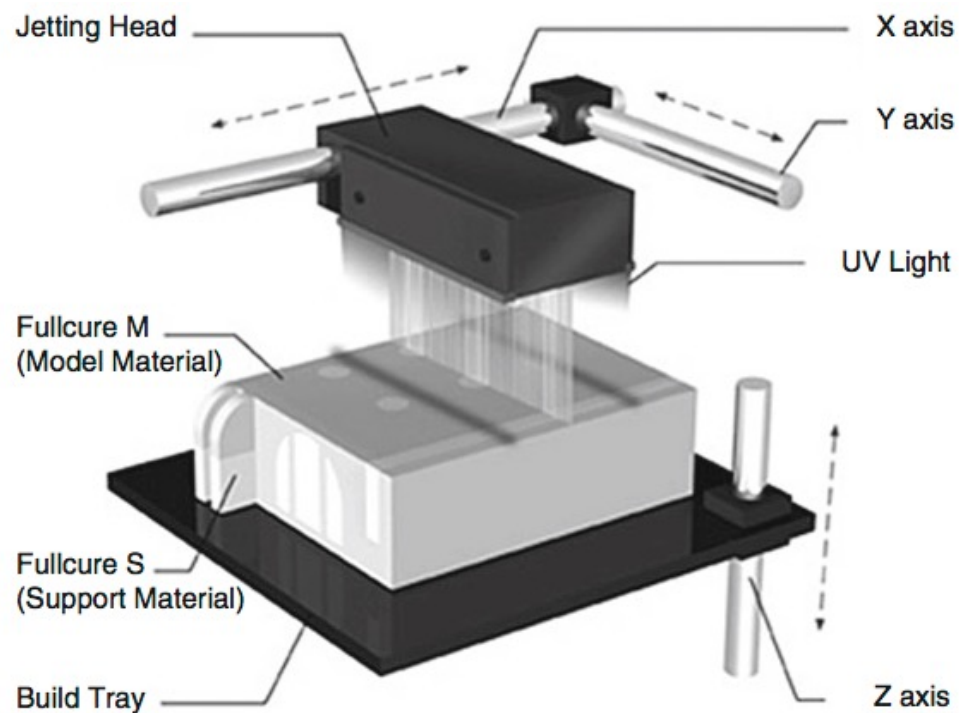
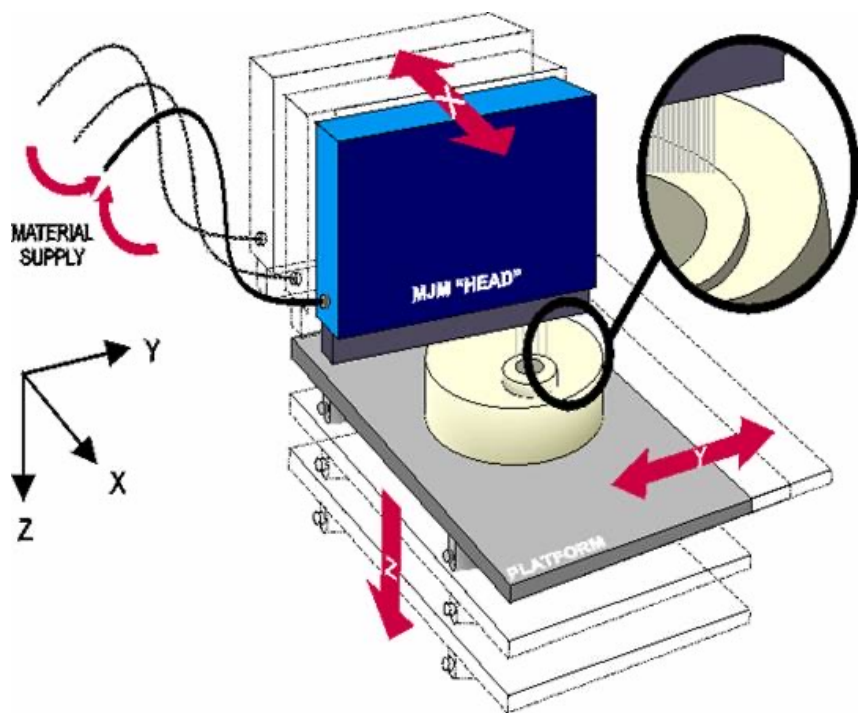


Printing technology has been extensively investigated, with the majority of that investigation historically based upon applications to the 2D printing industry (inkjet printing has been in existence since the 1960s).



Printing as a 3D building method was first demonstrated in the 1980s with patents related to the development of *Ballistic Particle Manufacturing*. The first commercially successful technology was introduced in 1994, which printed a basic wax material that was heated to liquid state.

It is notable that the first generation of 3D printing machines relied on heated **waxy thermoplastics** as their build material; they are therefore most appropriate for concept modeling and investment casting patterns. More recently, the focus of development has been on the deposition of **acrylate photopolymer**, wherein droplets of liquid monomer are formed and then exposed to ultraviolet light to promote photopolymerization.



Droplet Formation:

- Continuous Stream
- Drop on Demand



<https://youtu.be/kbildTVz6bA>



Primary advantages include:

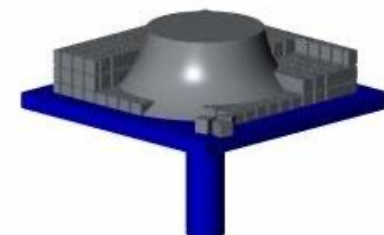
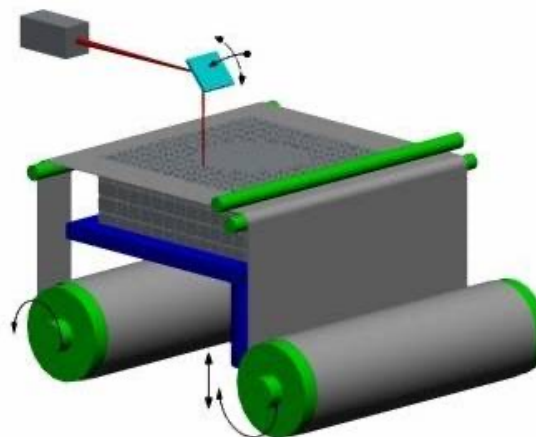
- low cost (no use of laser),
- high speed (form hundreds to thousands of nozzles),
- scalability (adding print heads),
- ease of building parts in multiple materials,
- capability of printing colors.



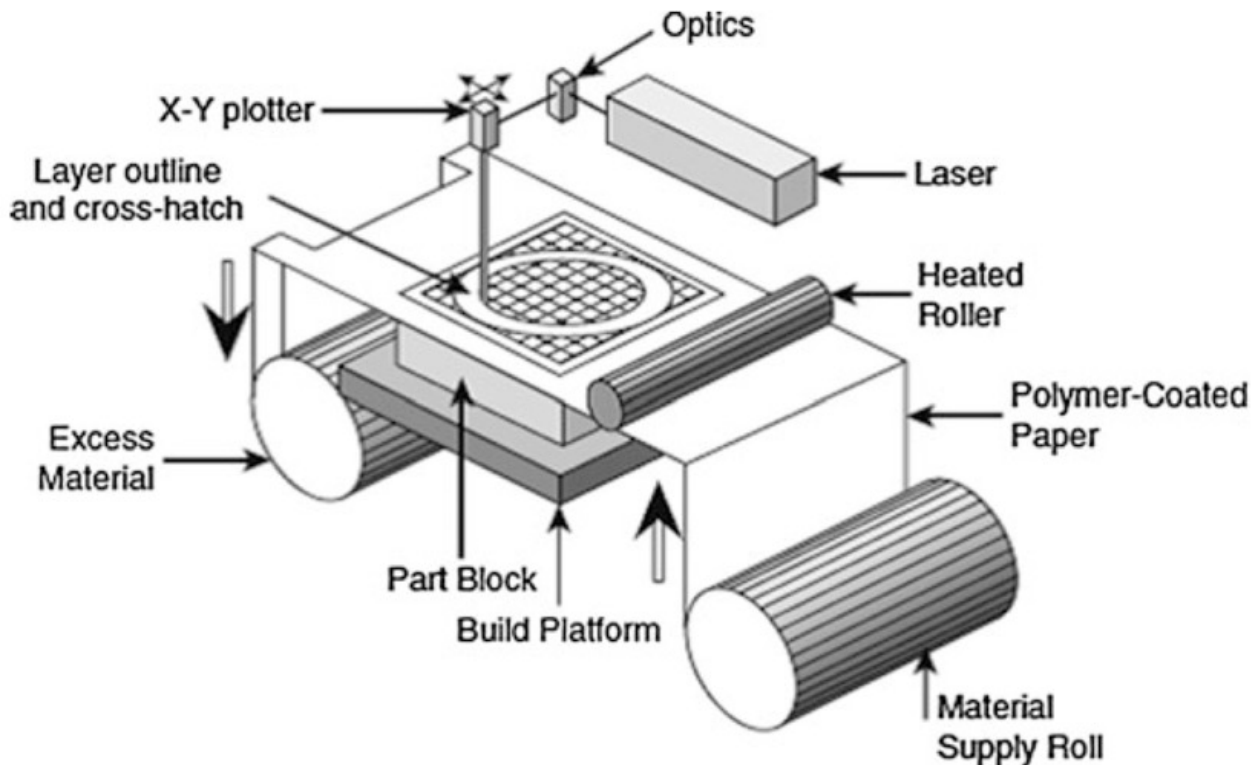
Sheet lamination: an additive manufacturing process in which sheets of material are bonded to form an object.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



One of the first commercialized (1991) AM techniques was Laminated Object Manufacturing (LOM).



LOM involved layer-by-layer lamination of paper material sheets, cut using a CO₂ laser, each sheet representing one cross-sectional layer of the CAD model of the part.

In LOM, the portion of the paper sheet which is not contained within the final part is sliced into cubes of material using a crosshatch cutting operation.



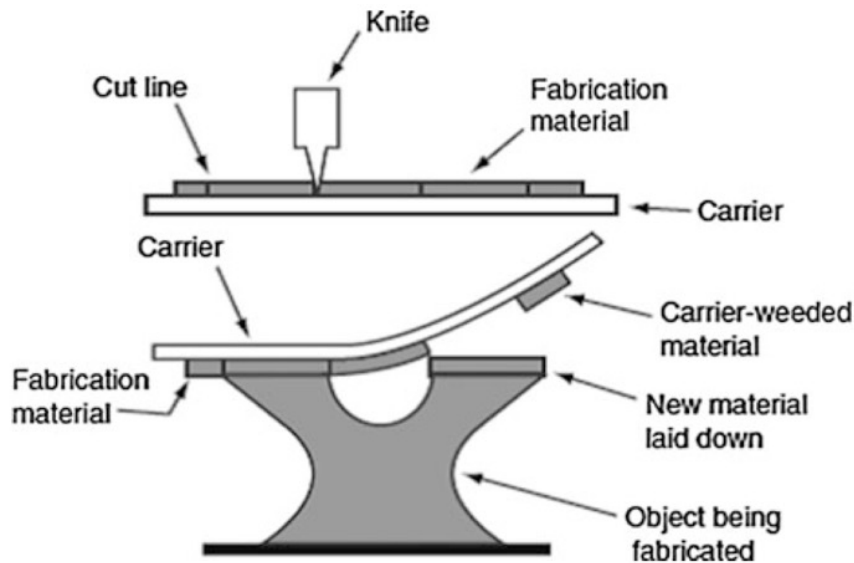
In some processes the laminate is bonded first to the substrate and is then formed into the cross-sectional shape (“**bond-then-form**” processes).



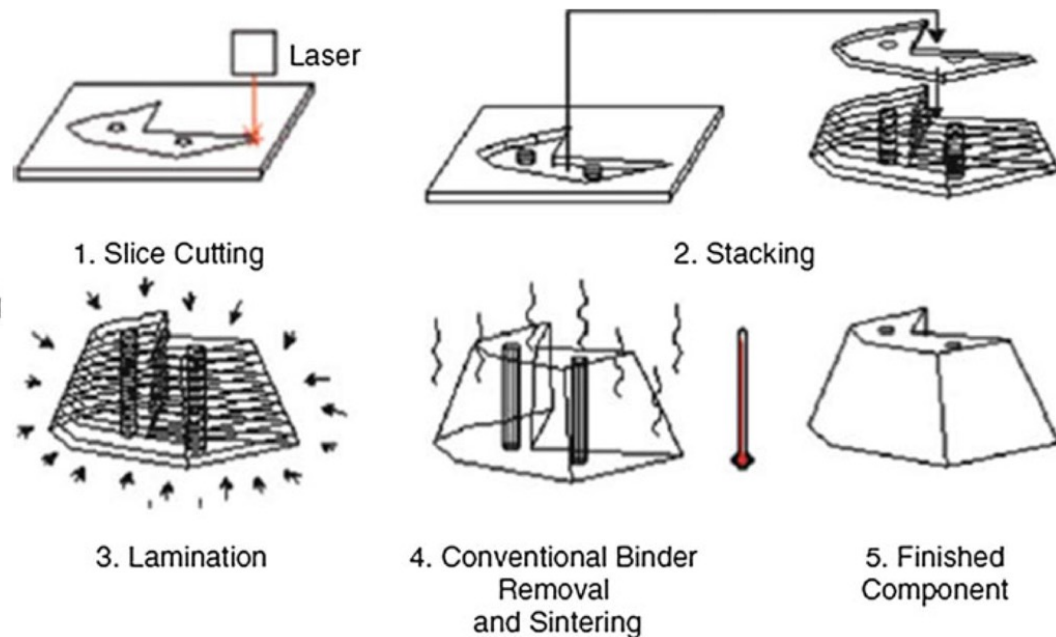
<https://youtu.be/aAbBvrfS6TY>



In **form-then-bond** processes, sheet material is cut to shape first and then bonded to the substrate. This approach is popular for construction of parts in metallic or ceramic materials that are thermally bonded.



Offset fabbing system

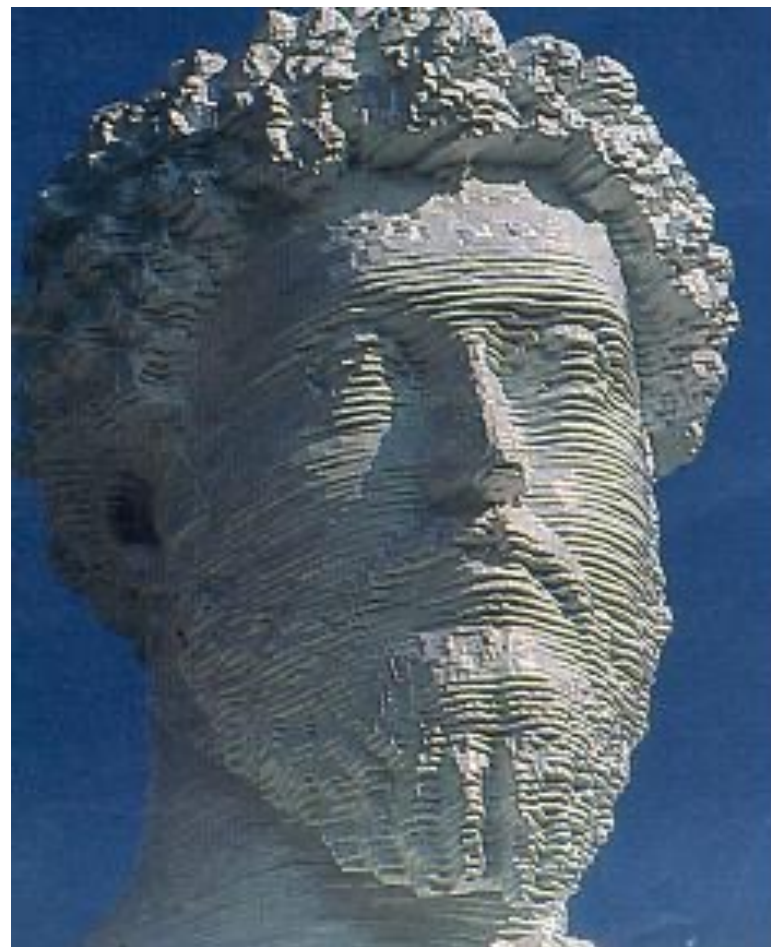


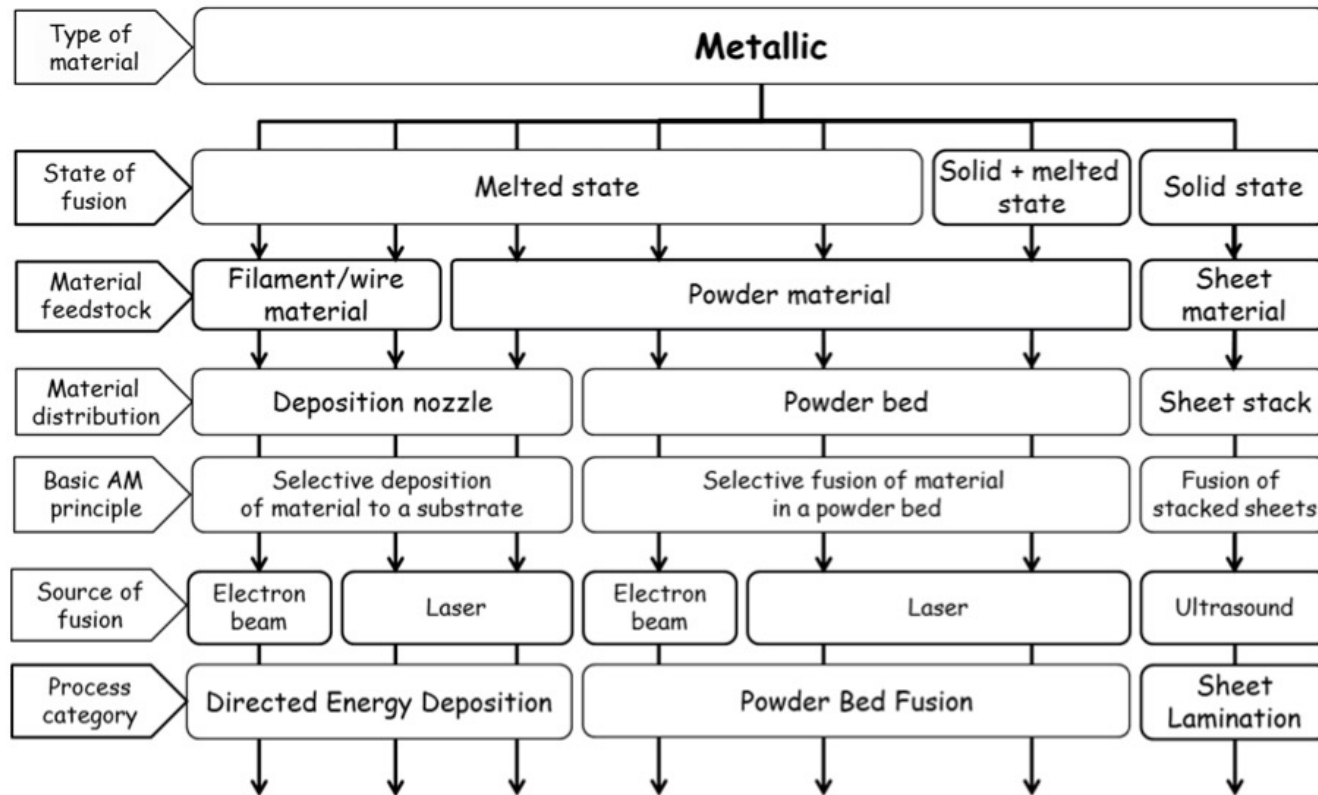
CAM-LEM Process

Computer-Aided Manufacturing of Laminated Engineering Materials



The Equestrian Statue of Marcus Aurelius
in the Campidoglio, Rome, Italy





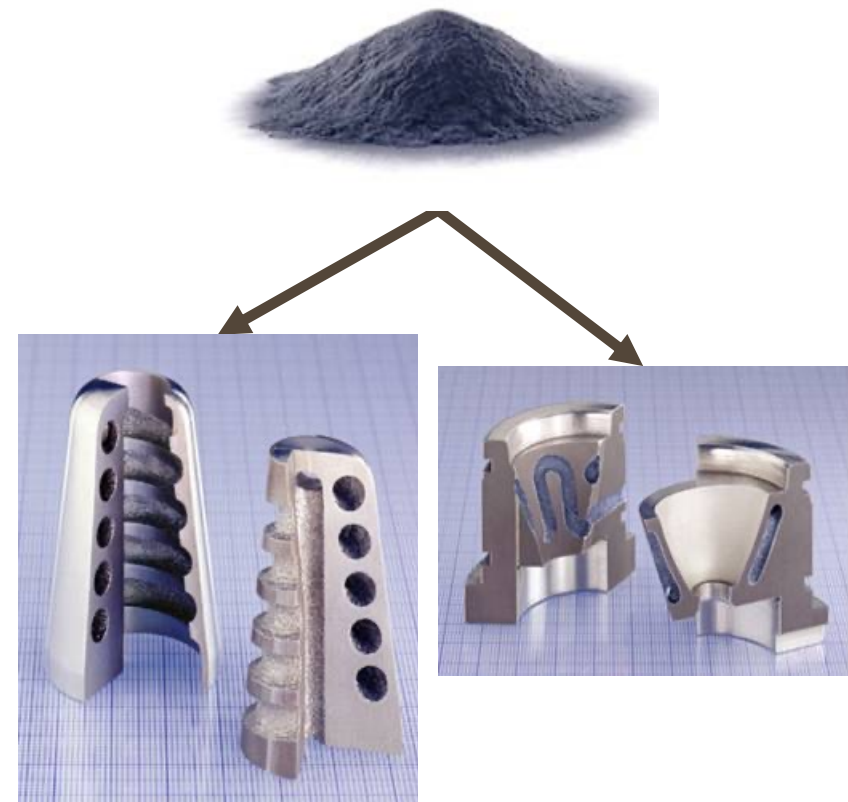


AM PROCESS CATEGORIES

Powder bed fusion: an additive manufacturing process in which thermal energy selectively fuses regions of a powder bed.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.





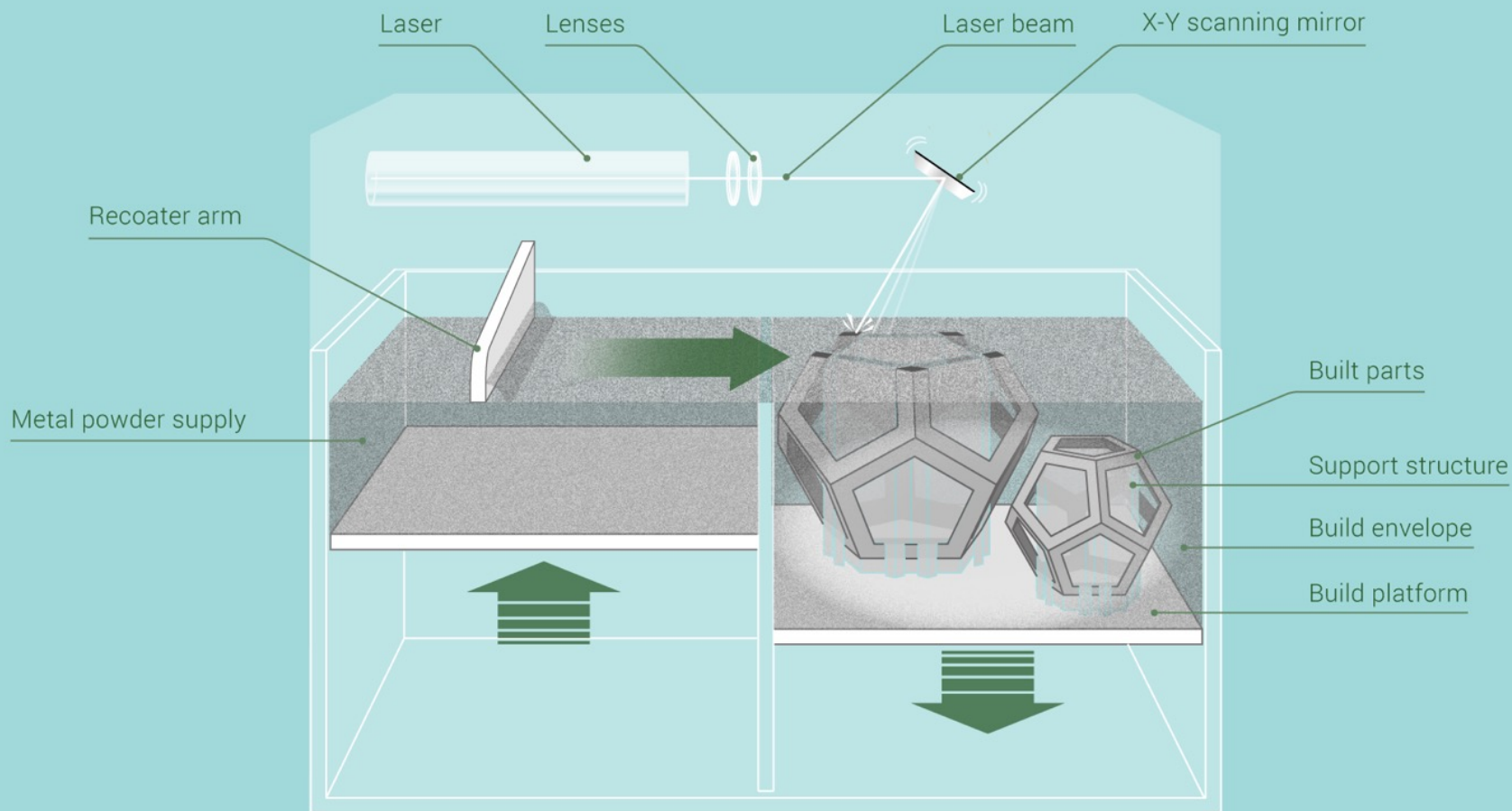
POWDER BED FUSION: PROCESSES

Then, according to the **energy source** and the **fusion mechanism**, there are:

- **Selective Laser Sintering (SLS):** Laser beam and liquid phase sintering
- **Selective Laser Melting (SLM):** Laser beam and full melting
- **Electron Beam Melting (EBM):** Electron beam and full melting



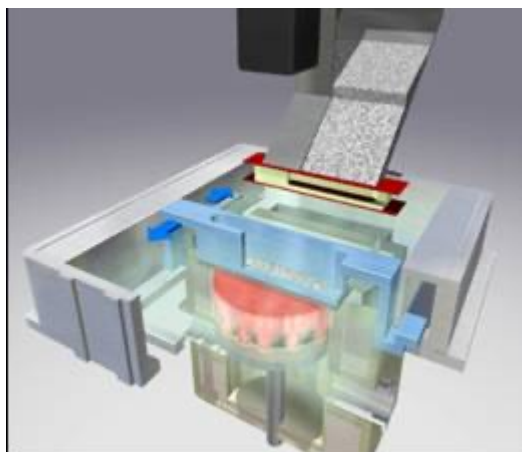
SELECTIVE LASER MELTING



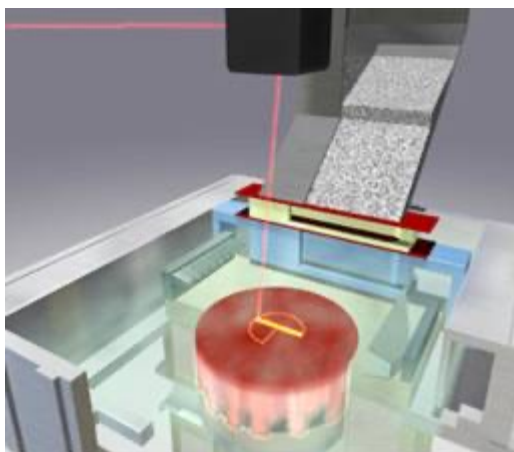
© **additively.com**



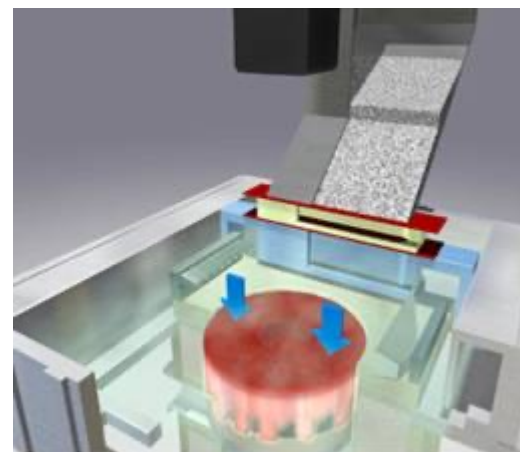
SELECTIVE LASER MELTING



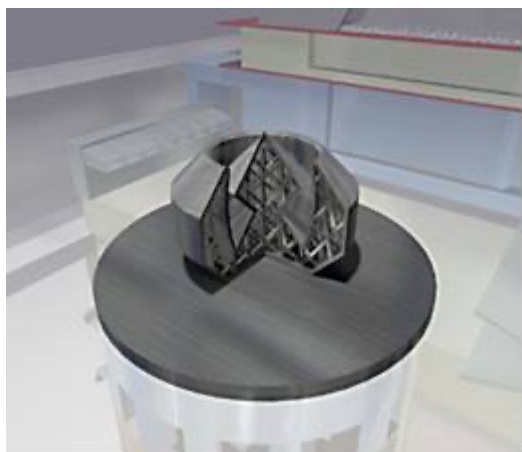
1) New powder layer



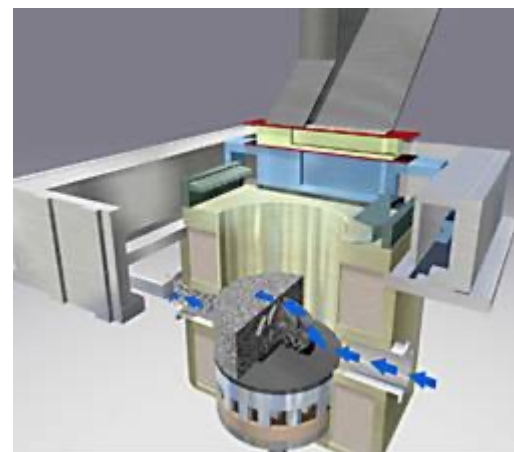
2) Powder melting



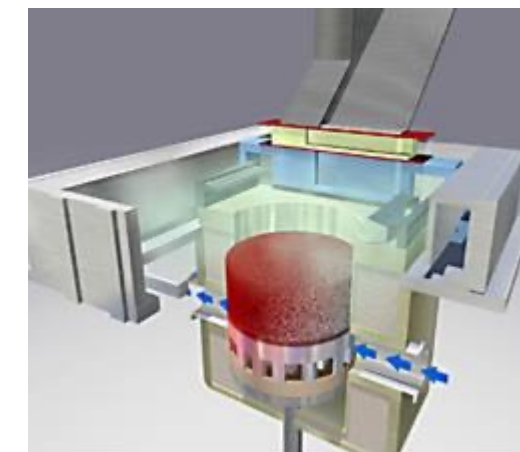
3) Platform lowering



6) Part



5) Extra powder removal



4) Cooling down

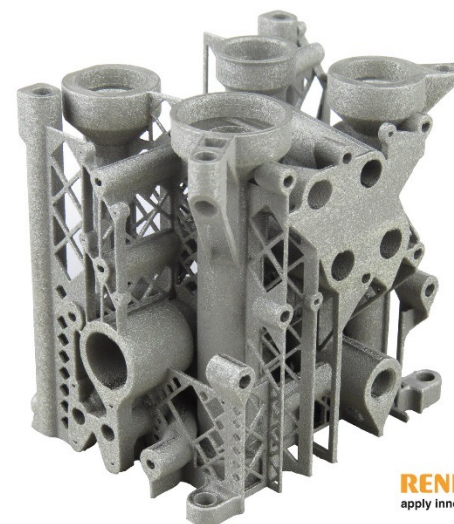


SELECTIVE LASER MELTING





SELECTIVE LASER MELTING



RENISHAW
apply innovation™

Development manifold produced on the Renishaw AM250 demonstrates how complex structures can be additively manufactured.

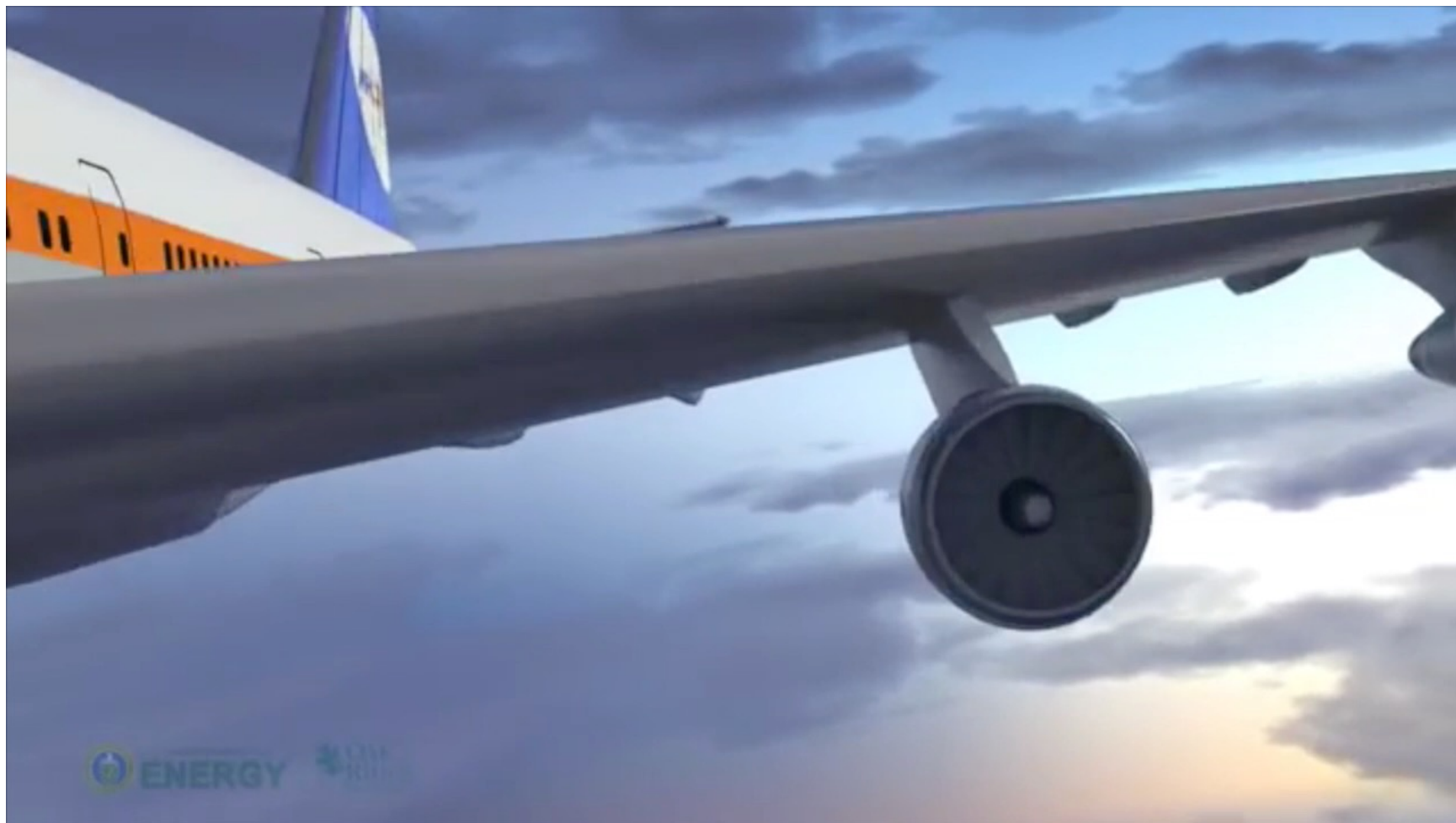
REF: REN013



Images courtesy of Renishaw



ELECTRON BEAM MELTING



Sorce: Arcam

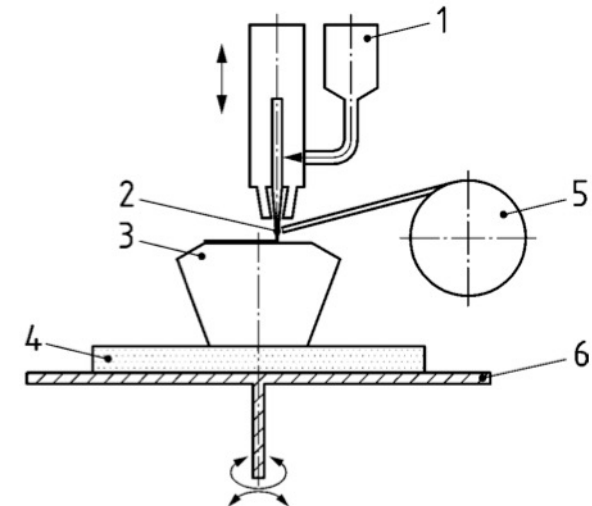


AM PROCESS CATEGORIES

Directed energy deposition (DED): an additive manufacturing process in which focused thermal energy is used to fuse materials by melting as they are being deposited.

Materials	Example materials	Process categories						
		Vat photo-polymerization	Material jetting	Binder jetting	Powder bed fusion	Material extrusion	Directed energy deposition	Sheet lamination
Thermoset Polymers	Epoxies and acrylates	X	X					
Thermoplastic polymers	Polyamide, ABS, PPSF		X	X	X	X		X
Wood	paper							X
Metals	Steel, Titanium alloys, Cobalt chromium			X	X		X	X
Industrial ceramic materials	Alumina, Zirconia, Silicone nitride	X		X	X			X
Structural ceramic materials	Cement, Foundry sand			X	X	X		

Note: Combinations of the above material classes, e.g. a composite, are possible.



Key

- | | |
|---|------------------------|
| 1 Powder hopper | 4 Substrate |
| 2 Directed energy beam, for example: laser, electron beam or plasma arc | 5 Wire (filament) coil |
| 3 Product | 6 Build table |



According to the **energy source** utilized one can distinguish:

- **EBMD – Electron Beam Metal Deposition**
- **LMD – Laser Metal Deposition**
- **PMD – Plasma Metal Deposition**

According to the **raw material type** one can distinguish:

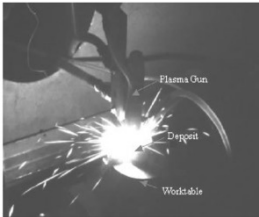
- **Powder Feeding**
- **Wire Feeding**

According to the **feeding modality** one can distinguish:

- **Coaxial Feeding**
- **Off-Axis Feeding**

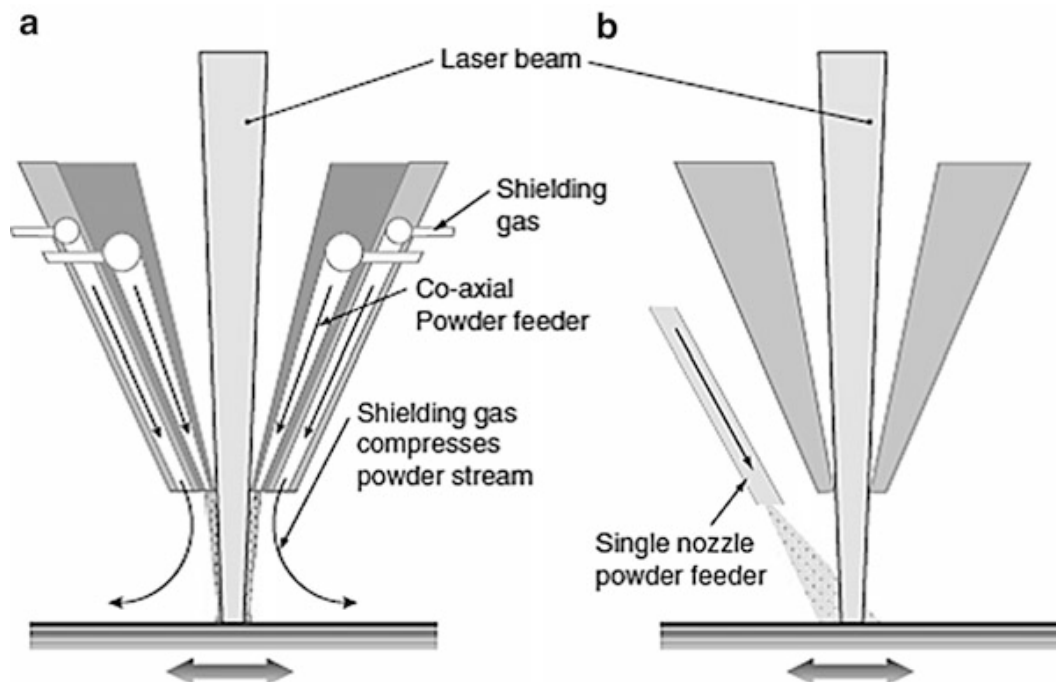


DIRECTED ENERGY DEPOSITION

	Powder	Wire
Laser	Laser metal deposition 	Laser wire deposition 
Electron beam	n/a	Electron beam wire AM 
Plasma/arc	Plasma deposition manuf. 	Plasma, arc wire feed AM 



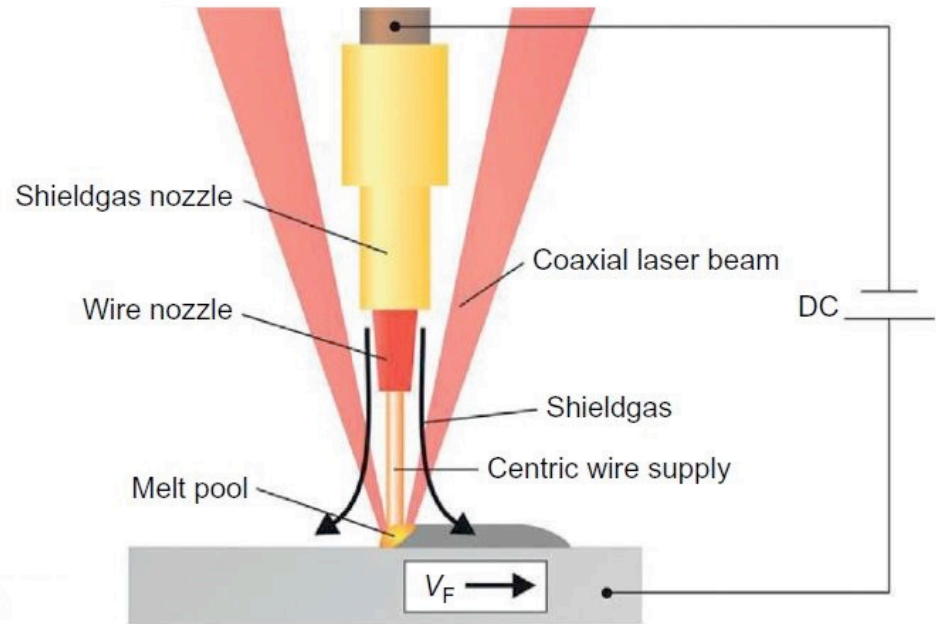
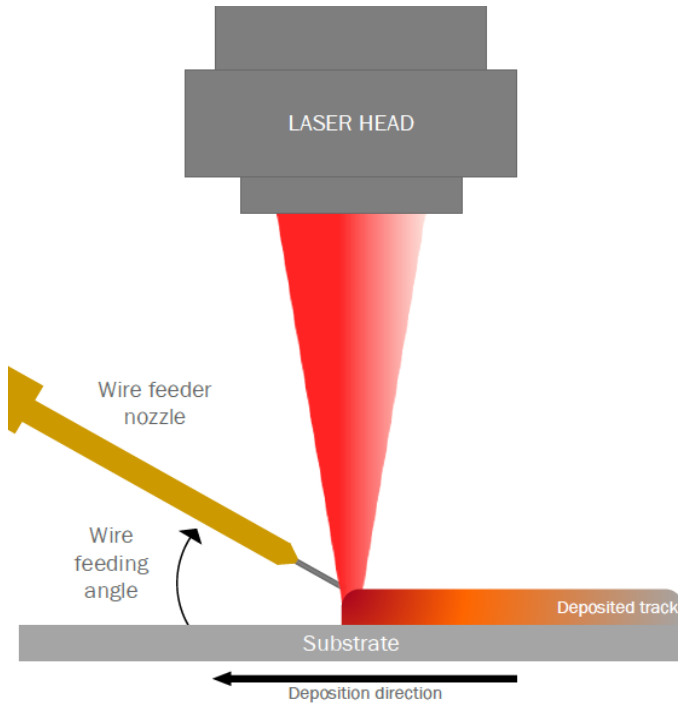
LASER METAL DEPOSITION: POWDER FEEDING



Coaxial feeding		Off-Axis feeding	
Pros	Cons	Pros	Cons
Higher flexibility	More complex nozzle design	Special nozzles are unnecessary	Less degrees of freedom
Symmetric deposition	More expensive head	Cheaper head	Non symmetric deposition



LASER METAL DEPOSITION: WIRE FEEDING



Off-Axis feeding		Coaxial feeding	
Pros	Cons	Pros	Cons
Cheaper Head	Non symmetric deposition	Symmetric deposition	Very complex nozzle design
Special nozzles are unnecessary	Less degrees of freedom	Higher flexibility	More expensive



POWDER BED FUSION VS DIRECTED ENERGY DEPOSITION

Assembly within build



High resolution



Powder
bed



Light-weight structures



Conformal cooling channels

High deposition rate



Large parts



Directed
energy



Improved material use



Build on existing parts



SAUER



SOME NEW INTERESTING DIRECTIONS

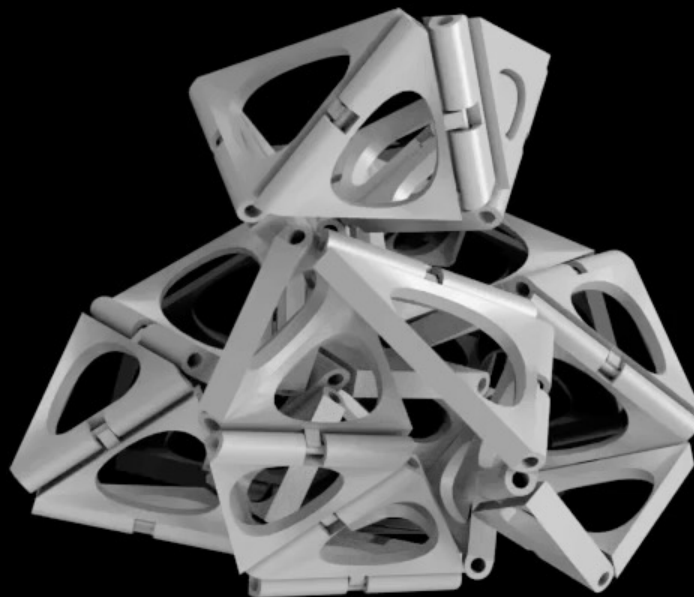
Camozzi e Politecnico di Milano insieme
per la ricerca e la crescita innovativa



<https://www.youtube.com/watch?v=k0poVtBhlsQ>



4D PRINTING



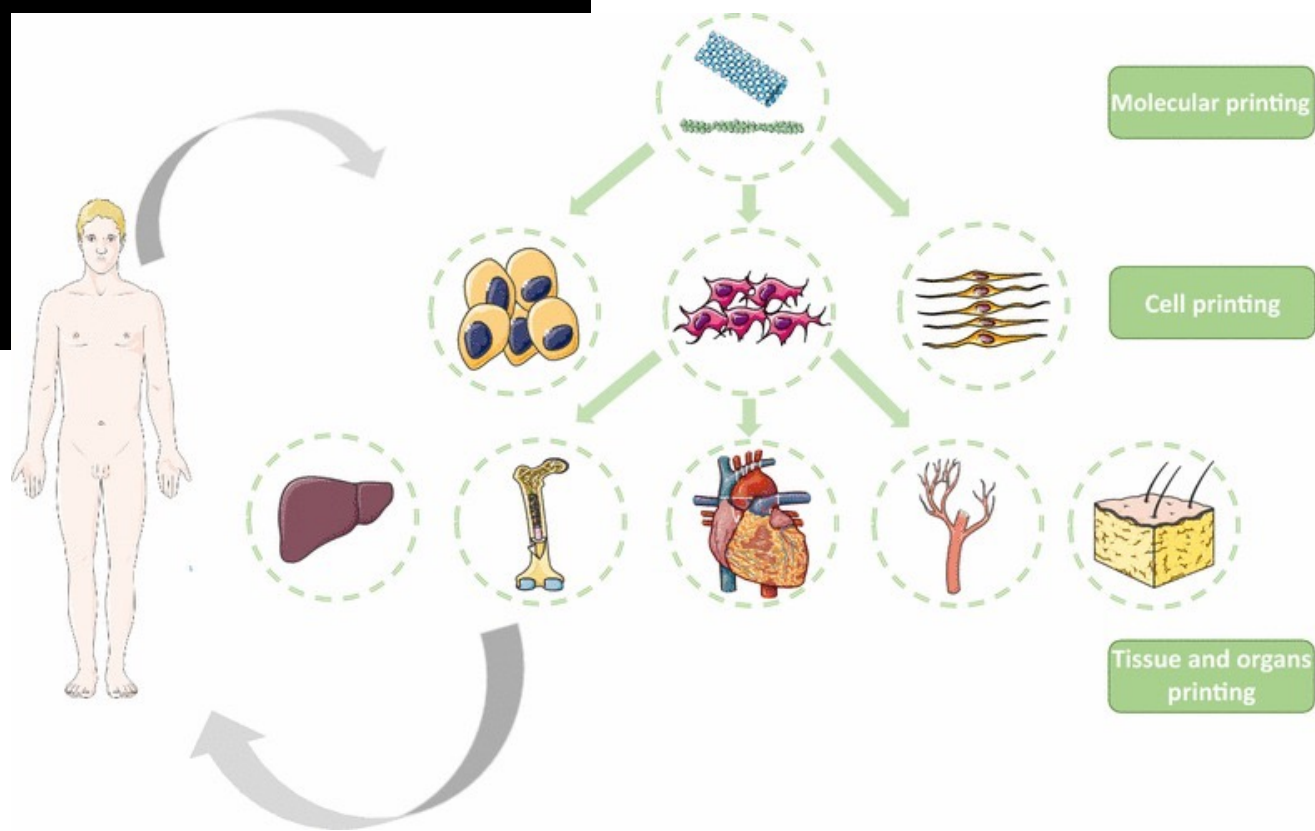


SOME NEW INTERESTING DIRECTIONS

62

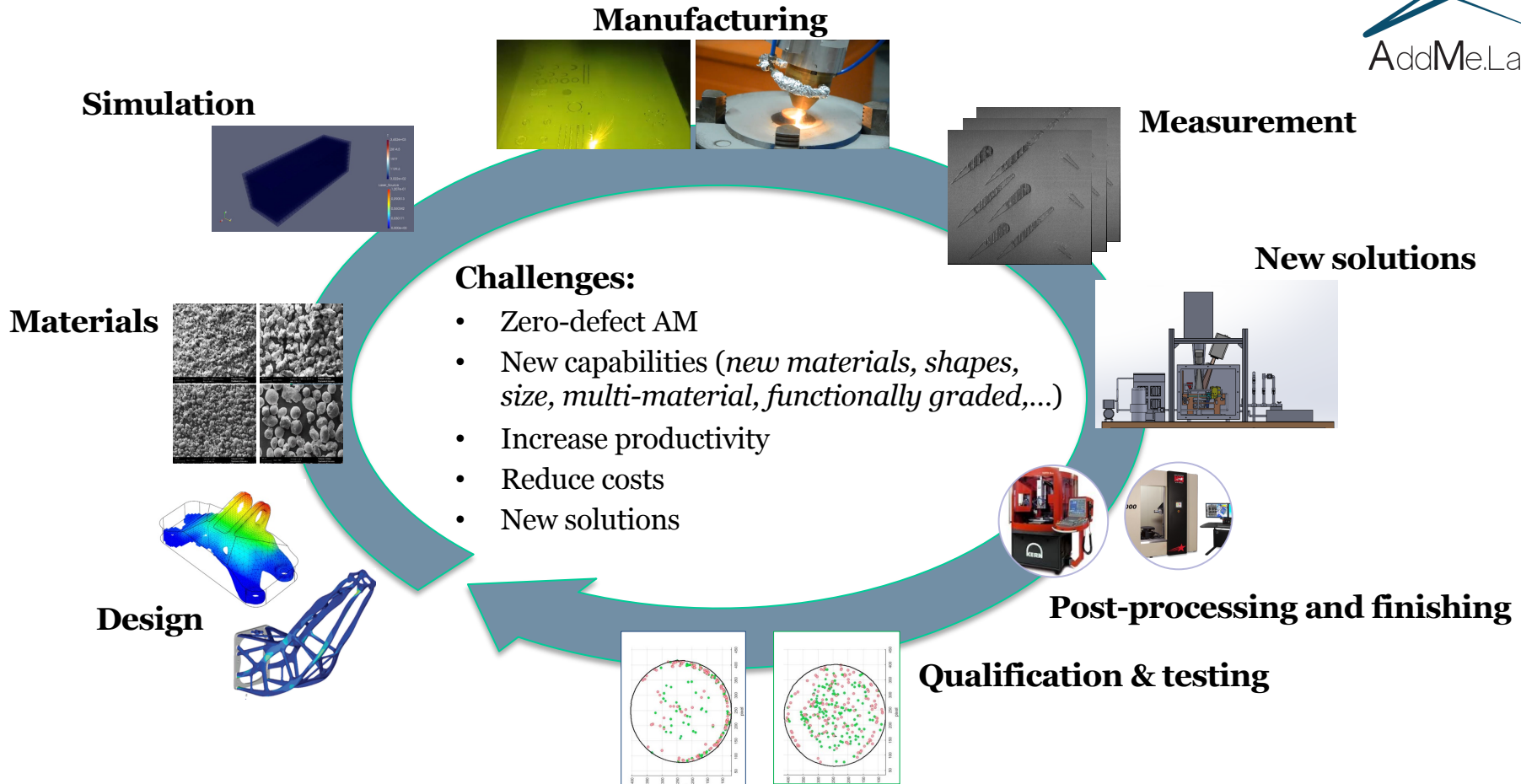


<http://www.3ders.org/articles/20160511-custom-made-pasta-italian-cuisine-giant-barilla-unveils-3d-pasta-printer.html>



https://www.youtube.com/watch?time_continue=8&v=s3CiJ26YS_U







METAL ADDITIVE MANUFACTURING @ MECHANICAL ENGINEERING DEPARTMENT

66

End users

THALES Ansaldo STS



Systems producers



Platforms/networks

Research institutions



Additive Manufacturing (5 CFU)
is among the courses of the TAB3 Group



prof. Bianca Maria Colosimo